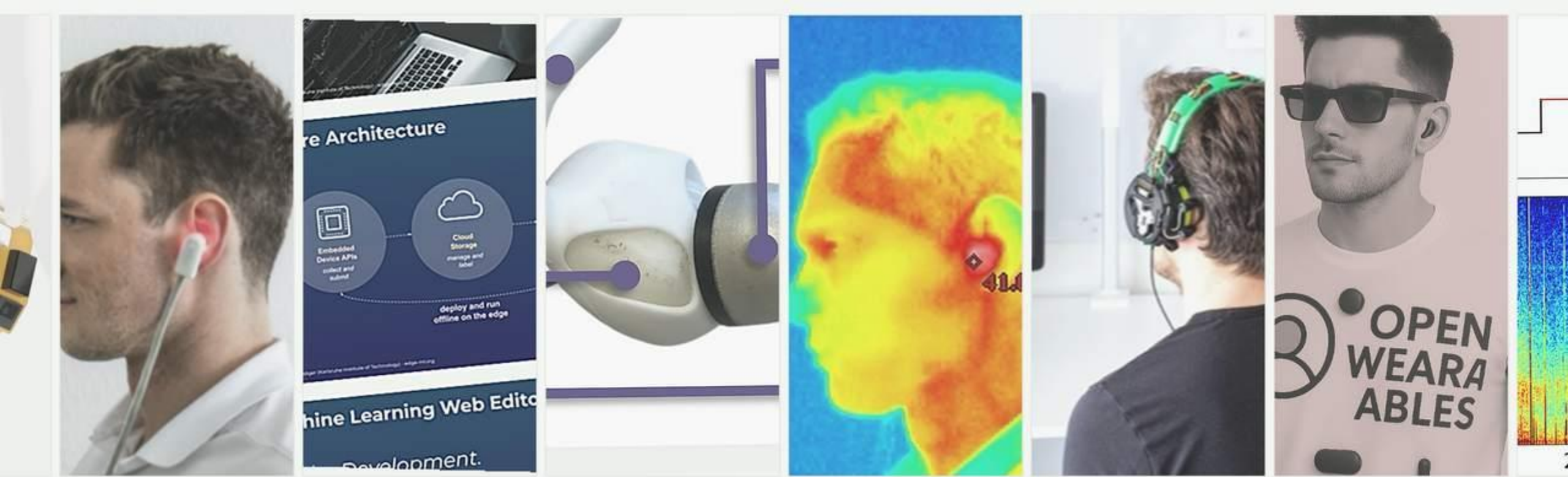




Human Computer Interaction SS26

Michael Beigl, TECO

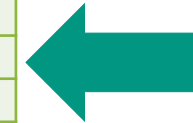
Chapter

3

Perception

Lecture Overview

Date / Day	Topic
21 Apr 2026 (Tuesday)	Introduction and Overview
22 Apr 2026 (Wednesday)	HIP
28 Apr 2026 (Tuesday)	HIP
29 Apr 2026 (Wednesday)	Exercise HIP / Lecture Integrated
05 May 2026 (Tuesday)	Perception
06 May 2026 (Wednesday)	Perception
12 May 2026 (Tuesday)	Design Analysis
13 May 2026 (Wednesday)	Exercise Design Analysis and Perception
19 May 2026 (Tuesday)	Design Analysis
20 May 2026 (Wednesday)	Design
26 May 2026 (Tuesday)	Pentecost / no lecture
27 May 2026 (Wednesday)	Pentecost / no lecture
02 Jun 2026 (Tuesday)	no lecture
03 Jun 2026 (Wednesday)	Exercise Design Analysis
09 Jun 2026 (Tuesday)	Design
10 Jun 2026 (Wednesday)	Interface
16 Jun 2026 (Tuesday)	Interface
17 Jun 2026 (Wednesday)	Exercise Design and Interface
23 Jun 2026 (Tuesday)	Observation
24 Jun 2026 (Wednesday)	Observation
30 Jun 2026 (Tuesday)	Exercise Observation



Overview Chapter Perception



Human Perception

More info's (Links in Ilias):

- Benyon 3rd edition chapter 25: Perception and Navigation

Human Senses

Aristoteles five sense definition:

- Sight
- Hearing
- Touch
- Smell
- Taste

} This lecture focuses mostly on sight and hearing
(and maybe touch)

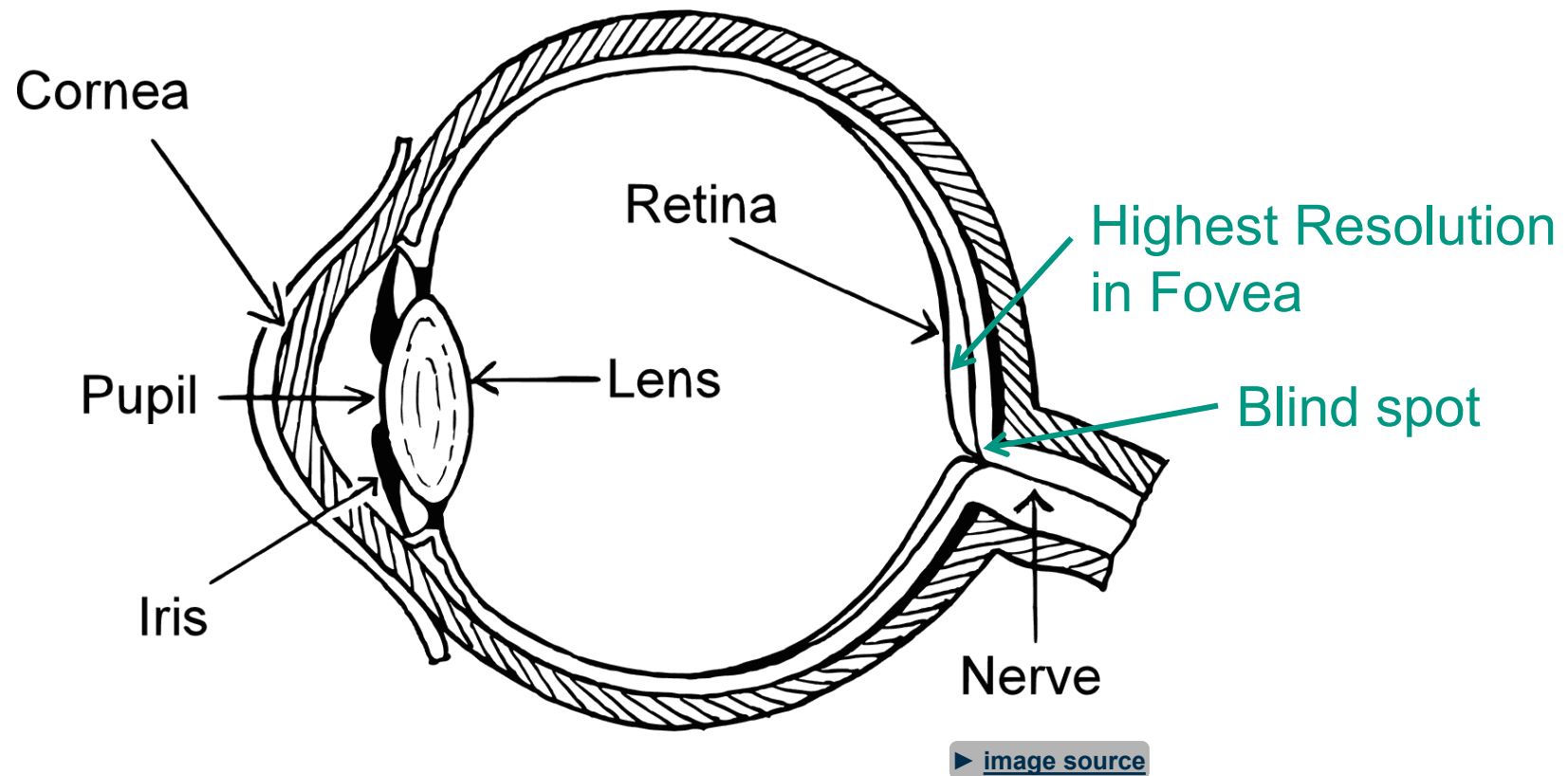


Additional senses

- nociception (pain),
- equilibrioception (balance),
- proprioception & kinesthesia (joint motion and acceleration),
- thermoception (temperature differences)

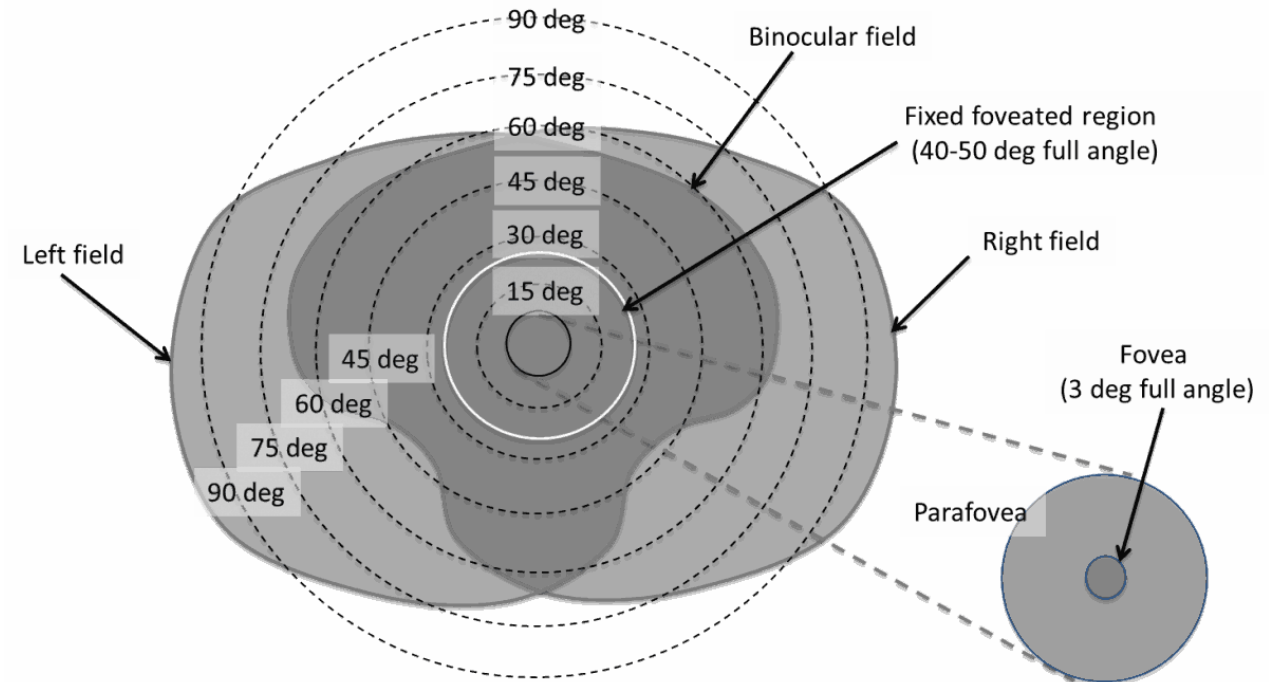
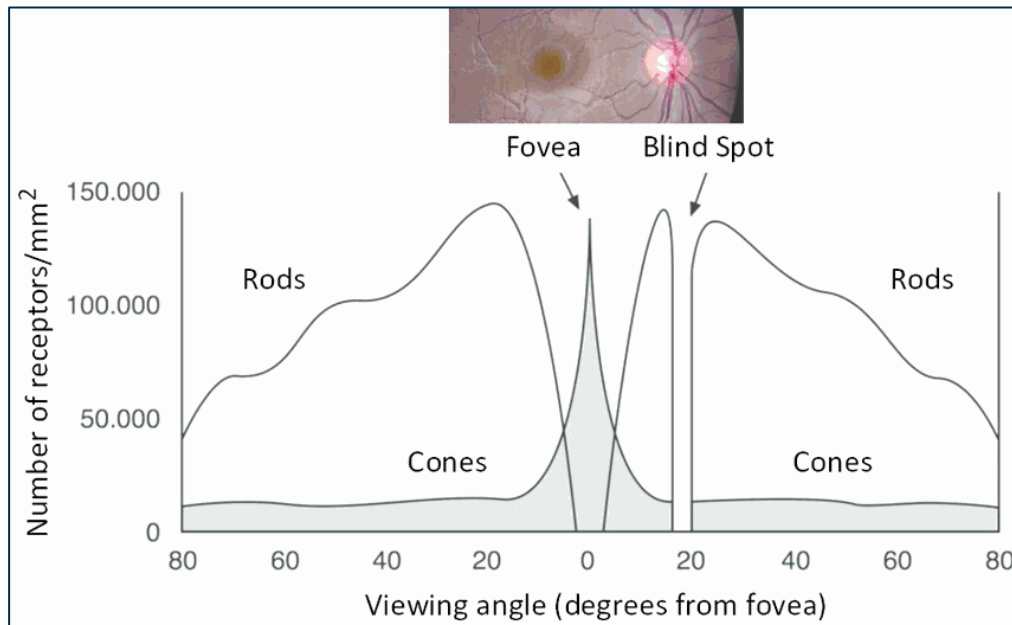
Final definition of senses open!

Human Eye



Human Eye

- **Cones**: Mainly in the middle of receptor cells (**colour**)
- **Rods**: Mostly on the outside (**black and white**)
- Very sharp area (foveal area): about 2°
- No blue colour receptors in the very middle



► [image source](#)

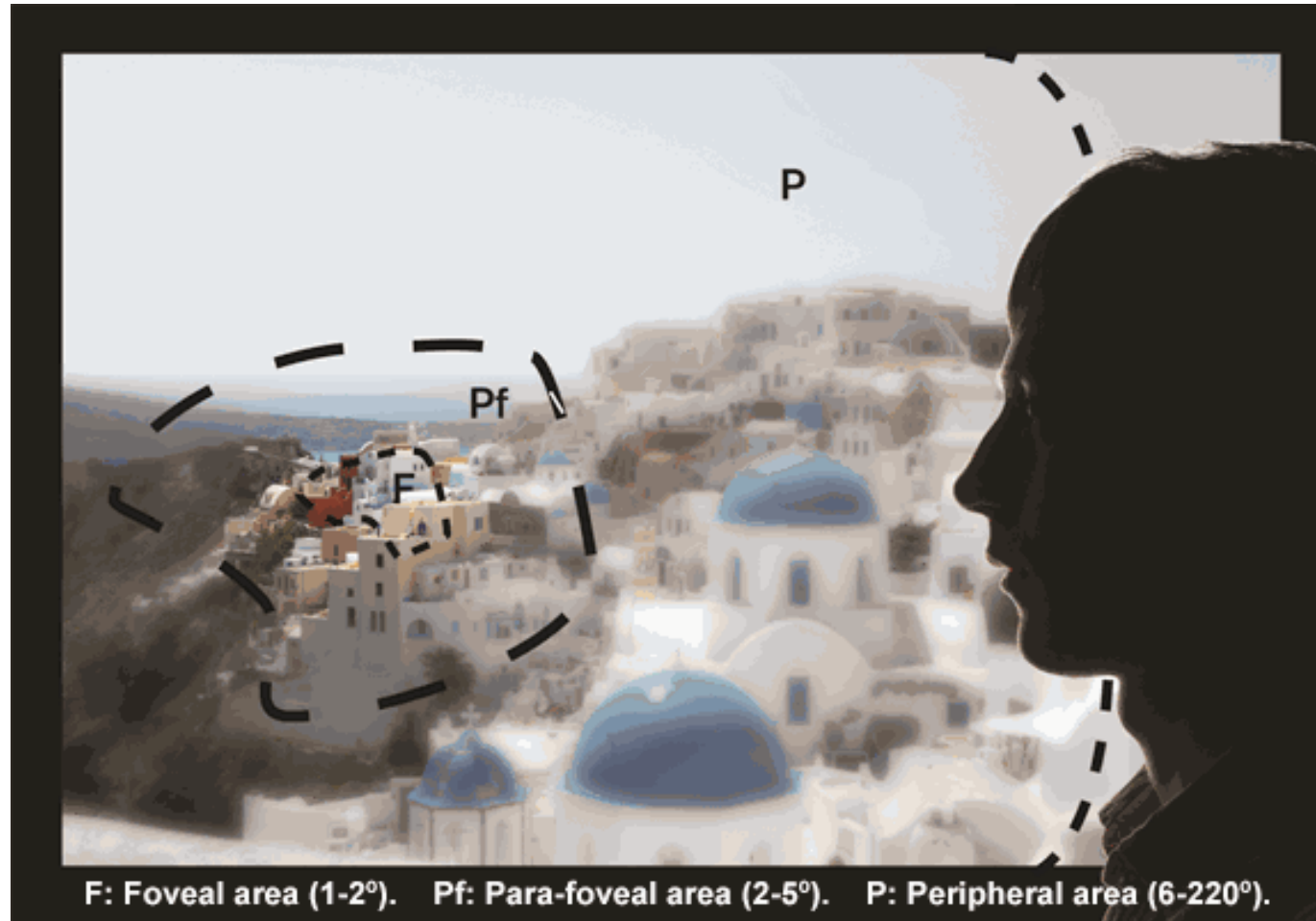
► [image source](#)

Visual sense – Some Facts



- More sharp in the middle of the focus
- Objects **nearer than 25cm** are difficult to focus
- Constant near-focusing leads to fatigue
- There is a **blind spot** about **15 – 20°** from the optical axis in your eye, so there are objects you are NOT able to see
 - But your eye moves constantly and your inner image is „recomputed“ using older images
- There is a maximum sampling frequency for images of about **22 images/sec**
- **Perception:** Humans are very sensitive to movement outside **about 30°** of the main focus
 - But are able to ignore: tunnel effect

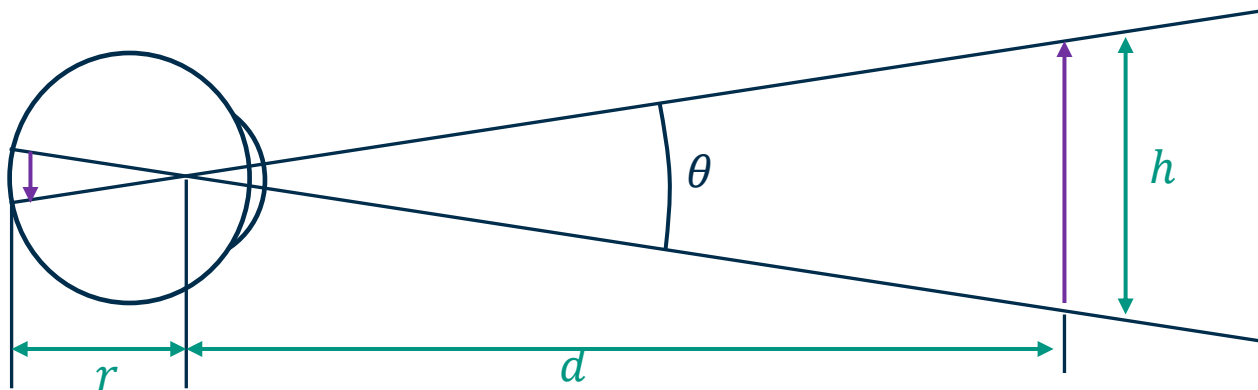
Foveal perception of a scene



[image source](#)

Reading distance vs. readable screen parts

- Mobile Phone, Tablet (standing, walking): 40 *cm*
- Watch: 50 *cm*
- Computer monitor: 55 –60 *cm*
- $h = 2d * \tan(\theta / 2)$
- $h = 14\text{--}35\text{ mm}$ → size Fovea/Para-Fovea ($2^\circ/5^\circ$) at 40 *cm* distance (mobile phone)
- $h = 21\text{--}52\text{ mm}$ → size Fovea/Para-Fovea ($2^\circ/5^\circ$) at 60 *cm* distance

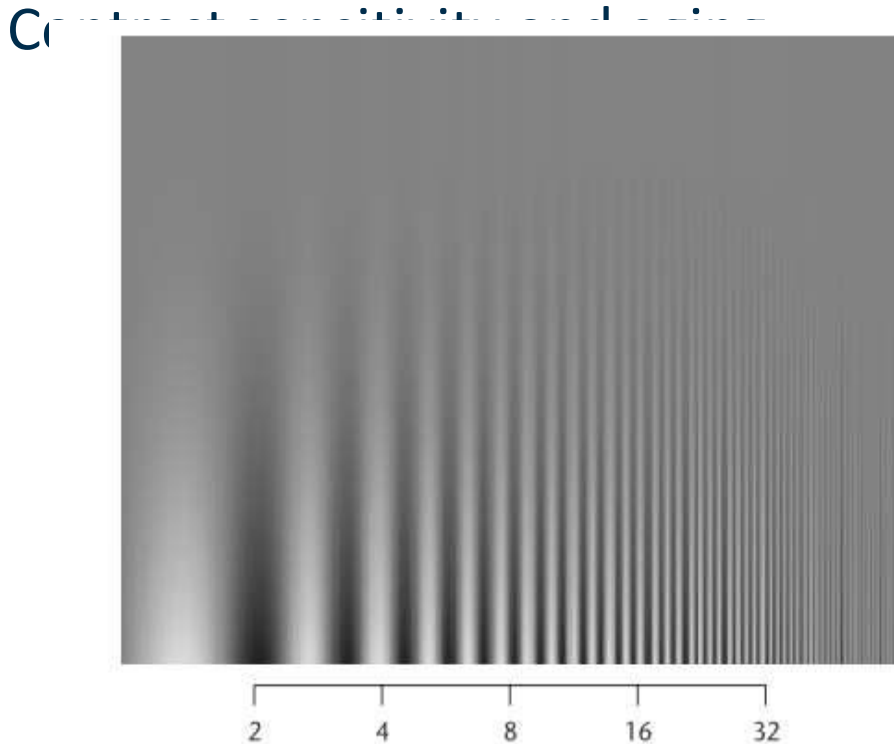


Presbyopia (Altersweitsichtigkeit)

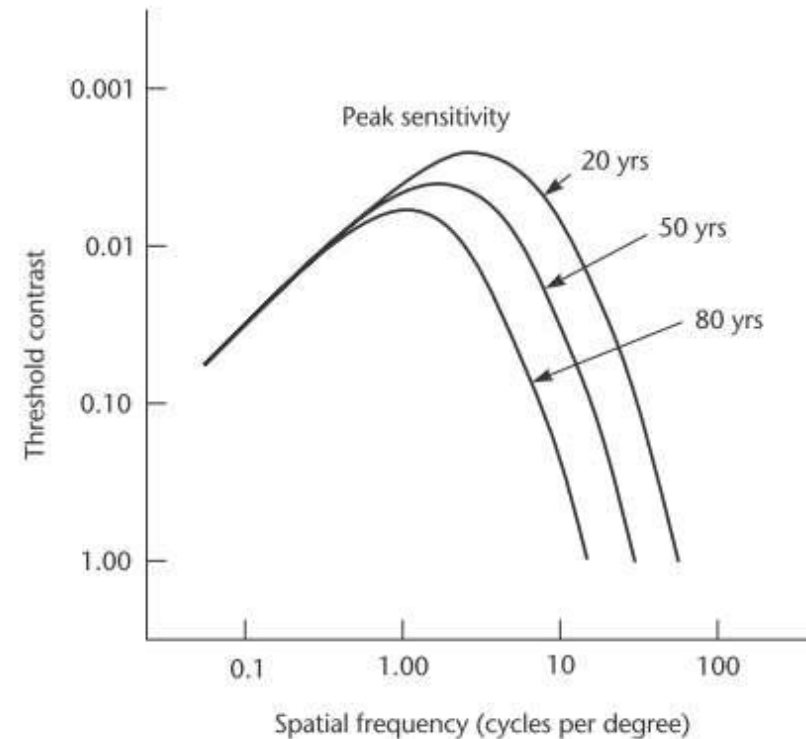
- Presbyopia happens to everybody and is unavoidable
- Avoid near point reading in general

Alter (Jahre)	Akkommodationsbreite (dpt)	Nahpunkt (cm)
10	13,5	7,5
20	10,0	10,0
30	7,5	13,5
40	4,5	22,0
45	3,5	28,5
50	2,5	40,0
55	1,5	66,5
60	1,0	100,0
65	0,5	200,0
70	0,25	400,0

► [image source](#)



This grating pattern changes frequency exponentially from left to right and varies in contrast in a vertical direction. The highest frequency you can resolve depends on the distance from which you view the pattern. The scale gives the spatial frequency if it is viewed from 2.3m.



Contrast sensitivity varies with spatial frequency. The function is illustrated for three age groups. As we age, our sensitivity to higher spatial frequencies is reduced. *Redrawn from Owlsley et al. (1983).*

Source: Book, Title "Information Visualization", Authors: Colin Ware, Morgen Kaufmann

Question:

Answer question about “contrast and foveale perception”

Link:

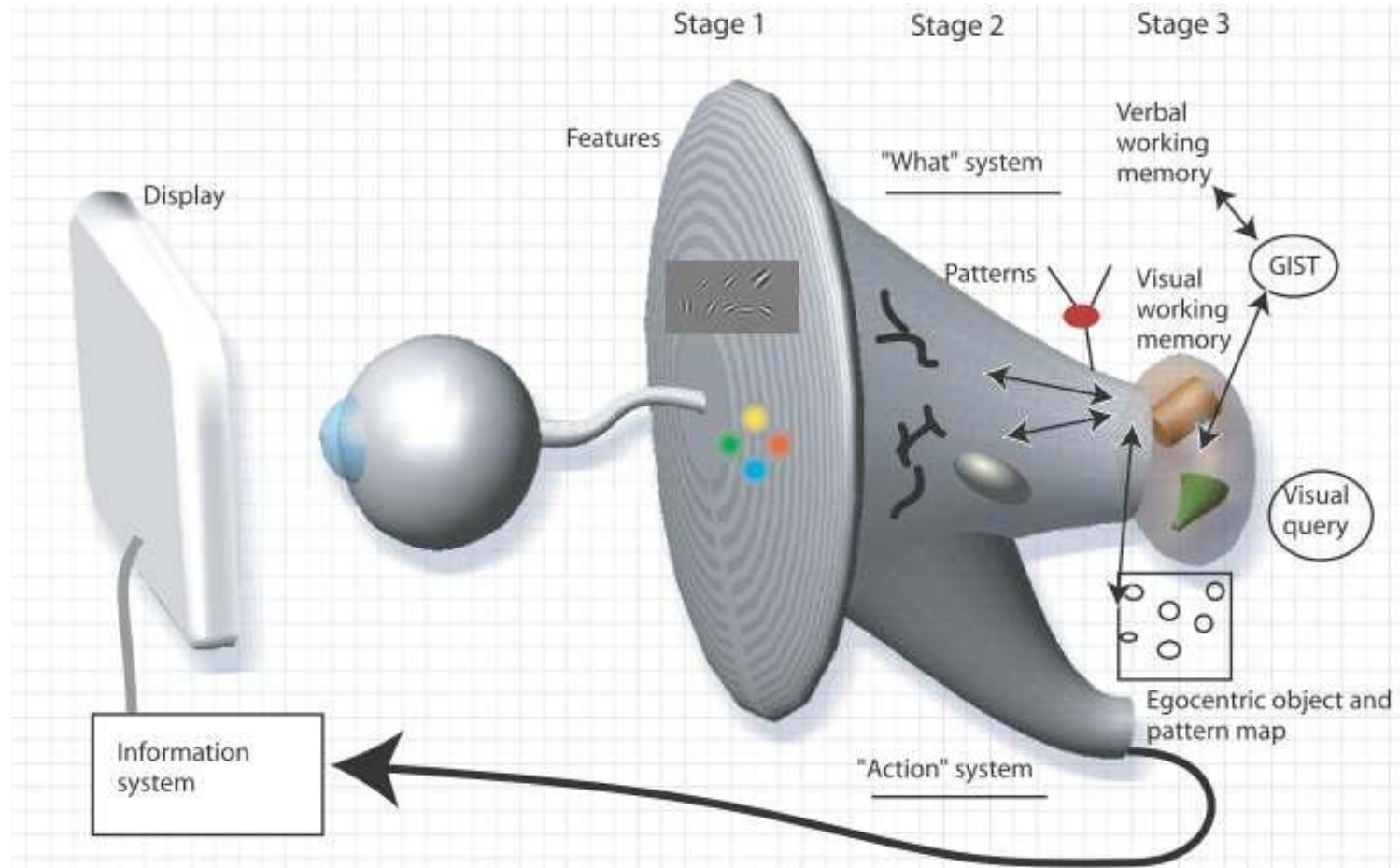
<https://pingo.coactum.de/589809>



Gestalt I Illusions



Visual Perception is about simplification



Source: Book, Title "Information Visualization", Authors: Colin Ware, Morgen Kaufmann

Foveale and cognitive perception heavy processing to the min.

Memory and „experience“ make up the cognitive image
→ this is e.g. also used to fill in the info for the blind spot



(a) Retinal image



(b) Ganglion layer



(c) Cortical Area

Simplifying to the “cartoony”

- This is why (good) cartoons of people can be immediately recognized
- ... and it can be even easier to process (if cartoon is matching the cortical areas)



(a) Retinal image



(b) Ganglion layer



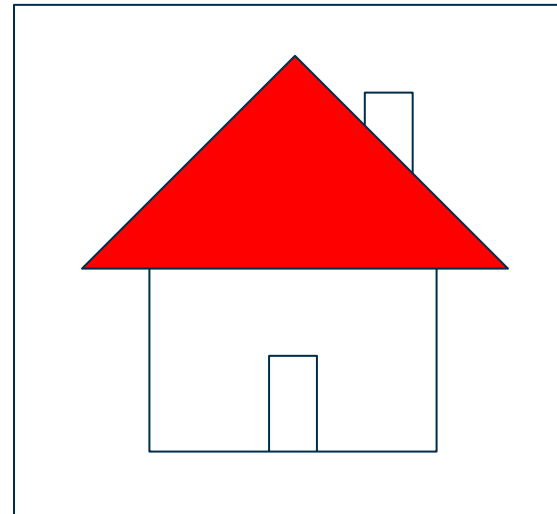
(c) Cortical Area



Visual Perception

■ **Visual perception** is the process of **extracting meaning** from sensory information. It is concerned with recognition and understanding.

■ **Vision** is an easier process concerned with **detecting color, shapes or edges of objects**. Vision does not necessary require an understanding of the world surrounding us.



Visual Perception:

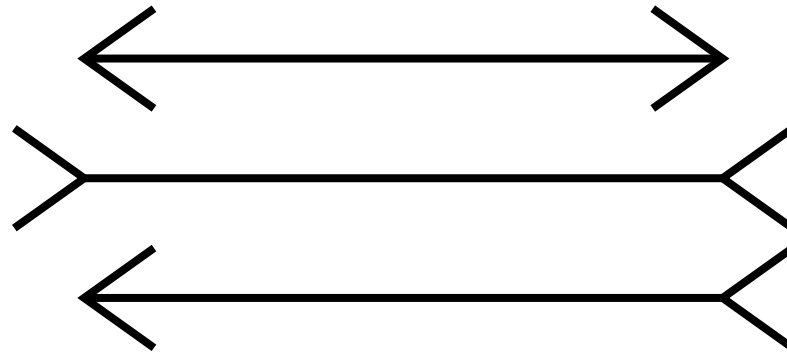
- house
- door
- roof ...

Vision:

- red triangular shape
- 2 rectangular shapes
- b/w lines ...

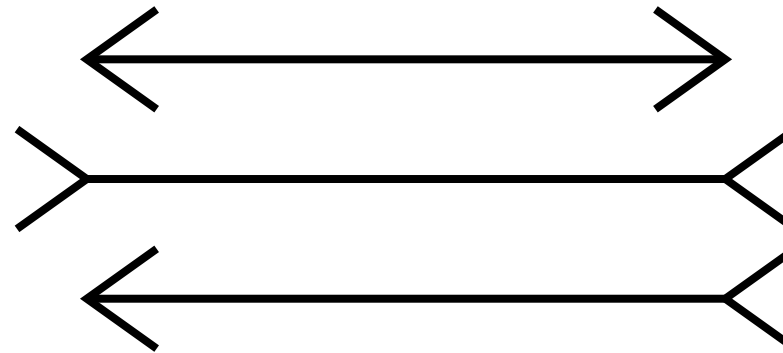
In Between: Patterns and Gestalt

Patterns: Visual Illusions – Müller-Lyer Illusion

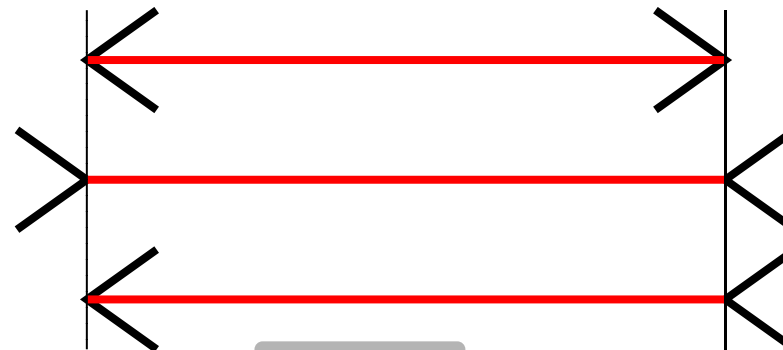


► [image source](#)

Visual Illusions – Müller-Lyer Illusion



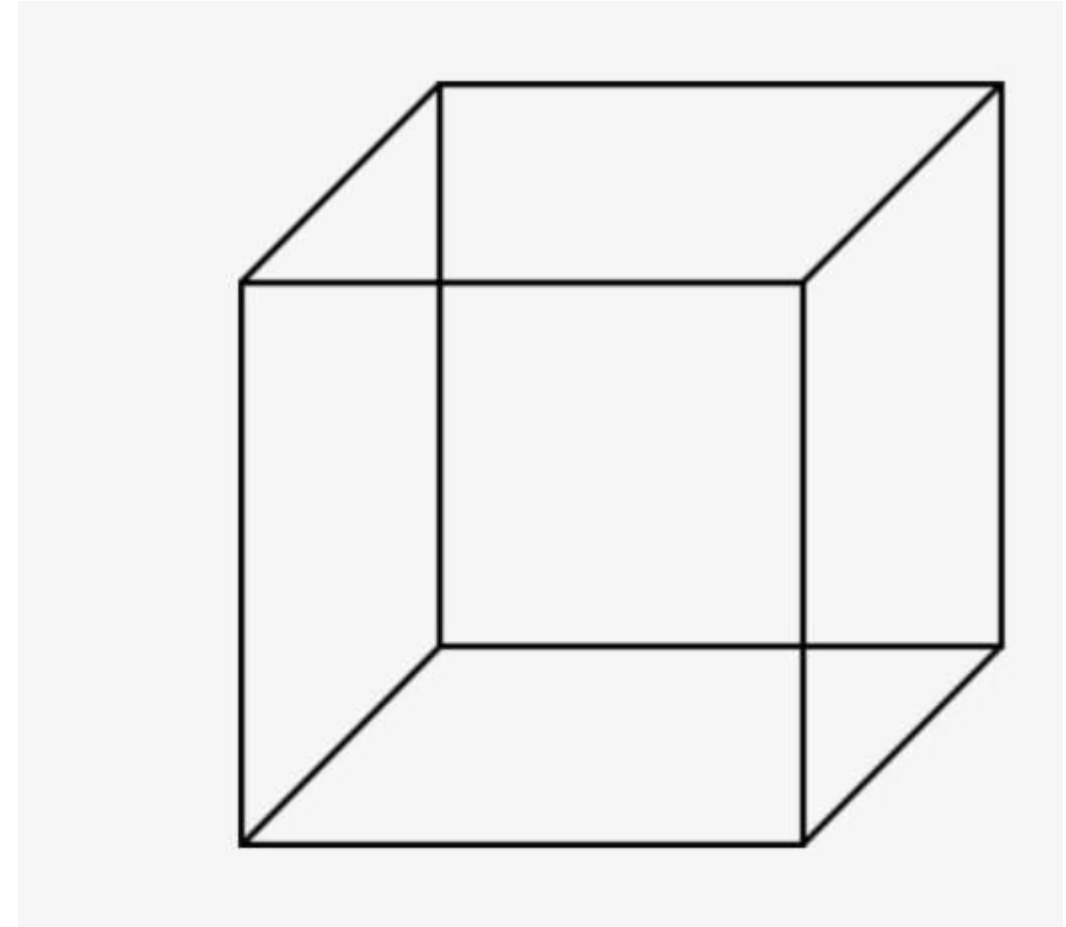
- All lines are equally long



► [image source](#)

Visual Illusions – Necker Cube

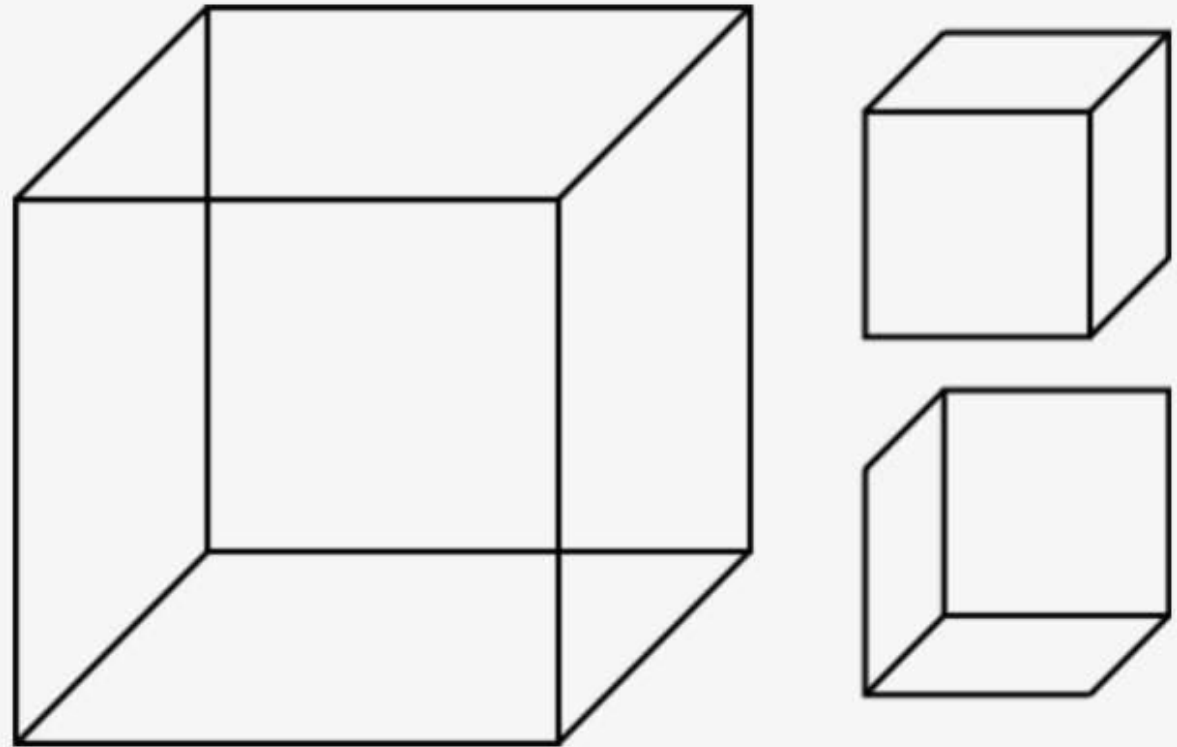
- Which cube do you see? There are two possible cubes.
- As our brain tries to make sense, both versions “swap”.



[▶ image source](#)

Visual Illusions – Necker Cube

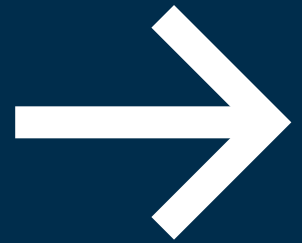
- Which cube do you see? There are two possible cubes.
- As our brain tries to make sense, both versions “swap”.



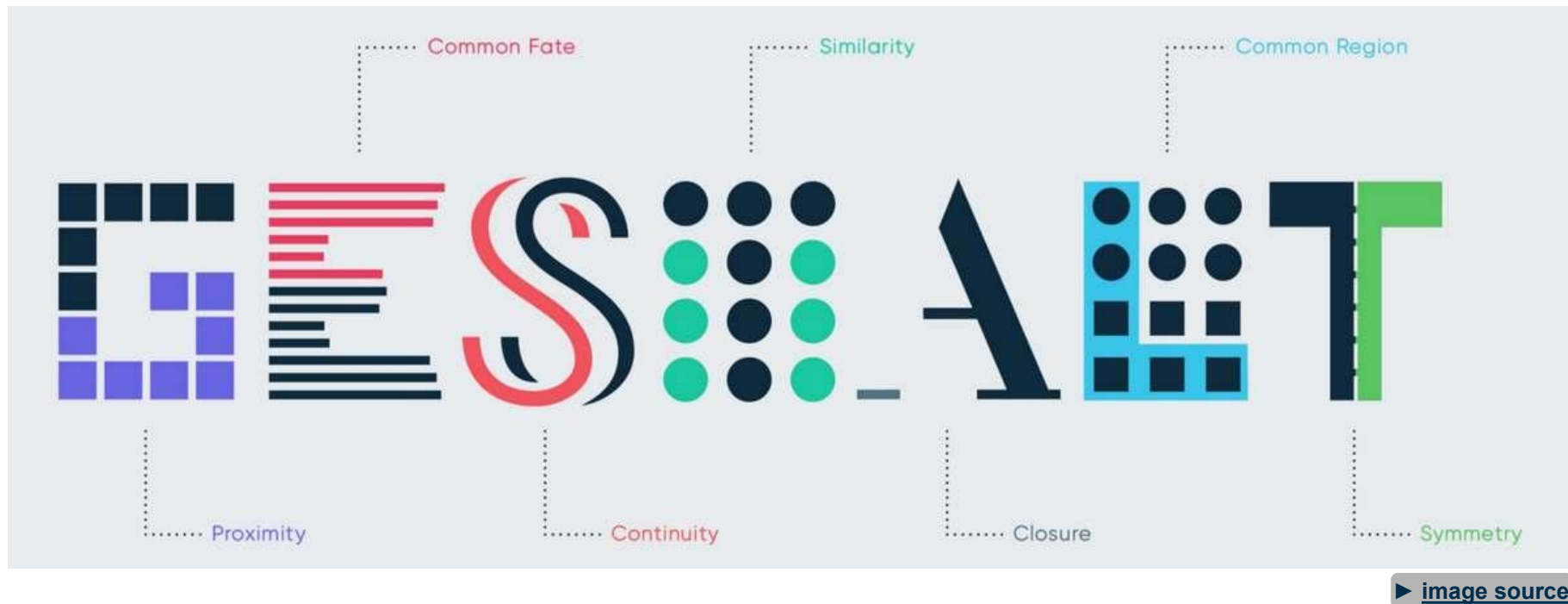
► [image source](#)

Gestalt II

Gestalt Laws

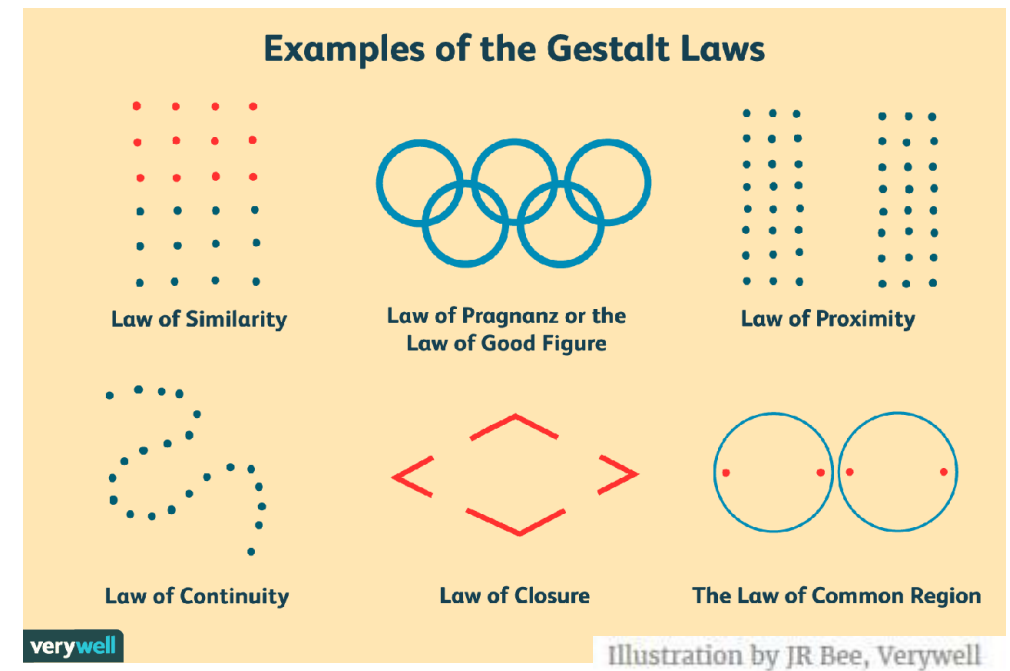


Gestalt Laws



Gestalt Laws

- The **Gestaltists**, a group psychologists, identified a number of properties that can be regarded as innate to all humans. Thus, their findings are **called Gestalt Laws**.
- Gestalt laws are methods that the brain uses to simplify recognition by ordering them
- **Gestalt Laws:**
 - Proximity
 - closure / Prägnanz
 - continuity / common fate
 - similarity / symmetry / Part-Whole Relationship
 - Area / Surroundness / Common Region

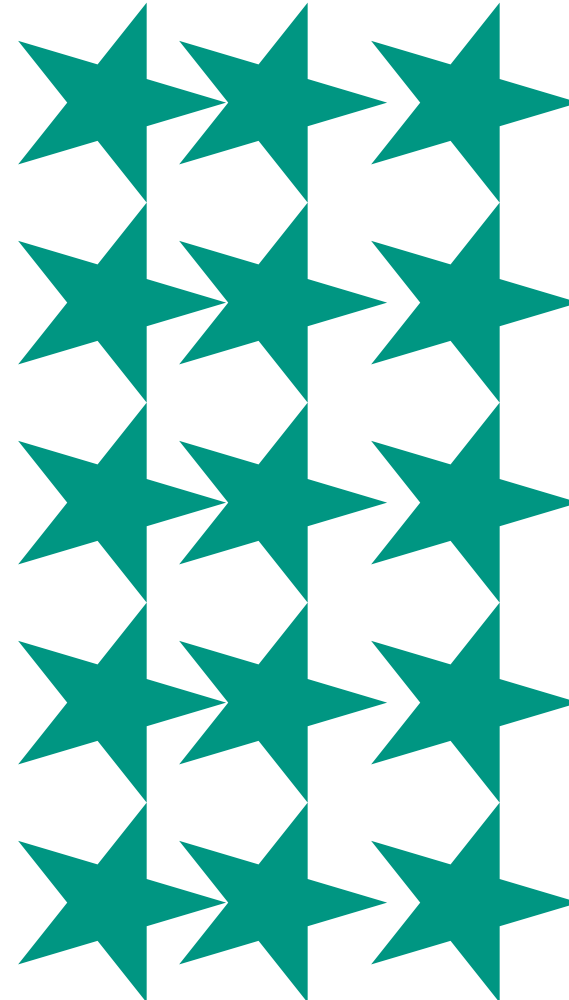


Designing Interactive Systems, D. Benyon, P. Turner and S. Turner, 2005, Addison-Wesley, p. 114-119

Gestalt Law – Proximity I

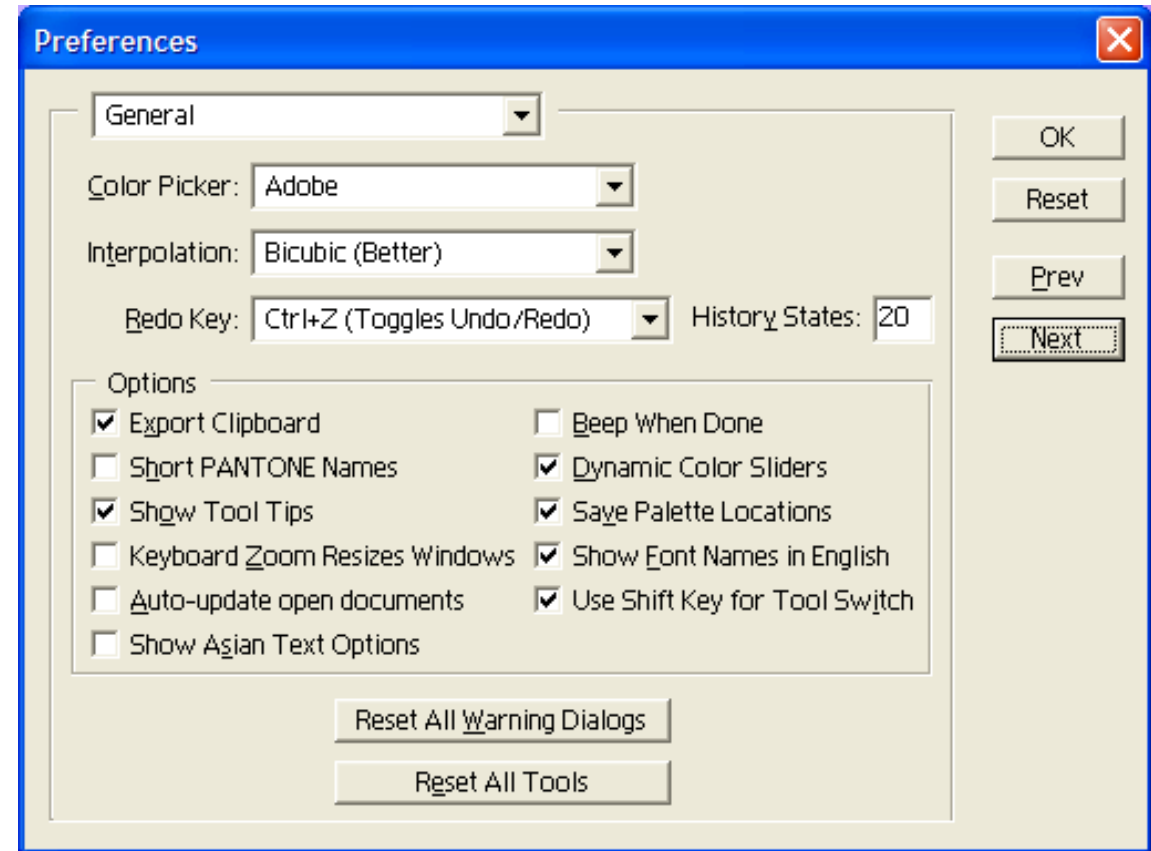


- Objects that are close in space or time tend to be perceived together.
- This can be used e.g. for UI arrangement of buttons or information

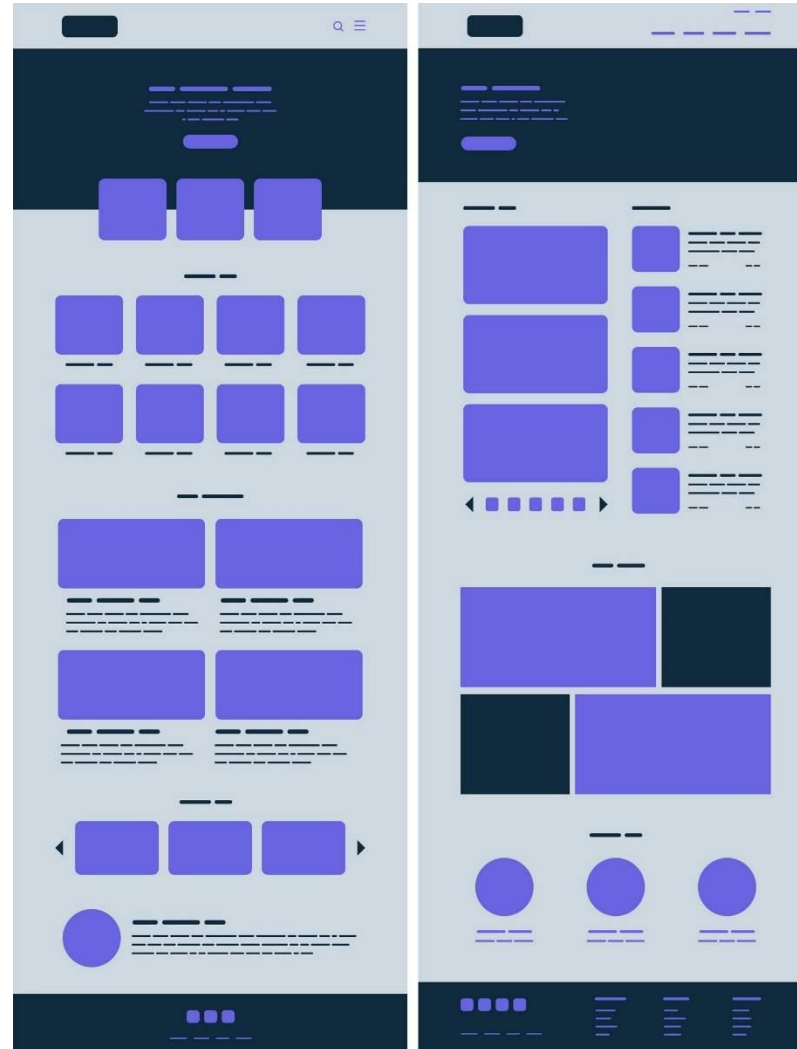


Gestalt Law – Proximity II

Proximity - Adobe PhotoShop Preferences Dialog



Proximity principles in UI

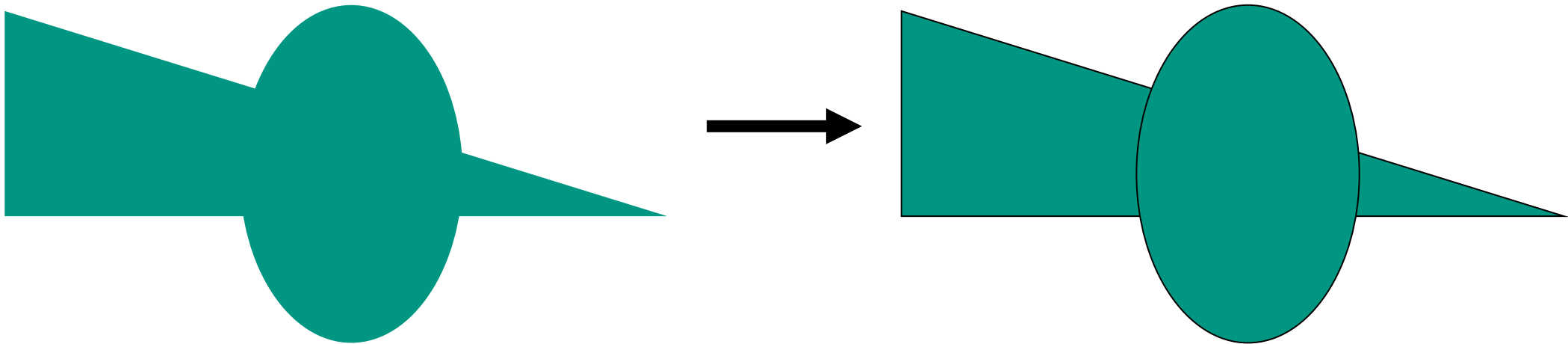


Prägnanz: Law of Good Gestalt

This is an unknown figure

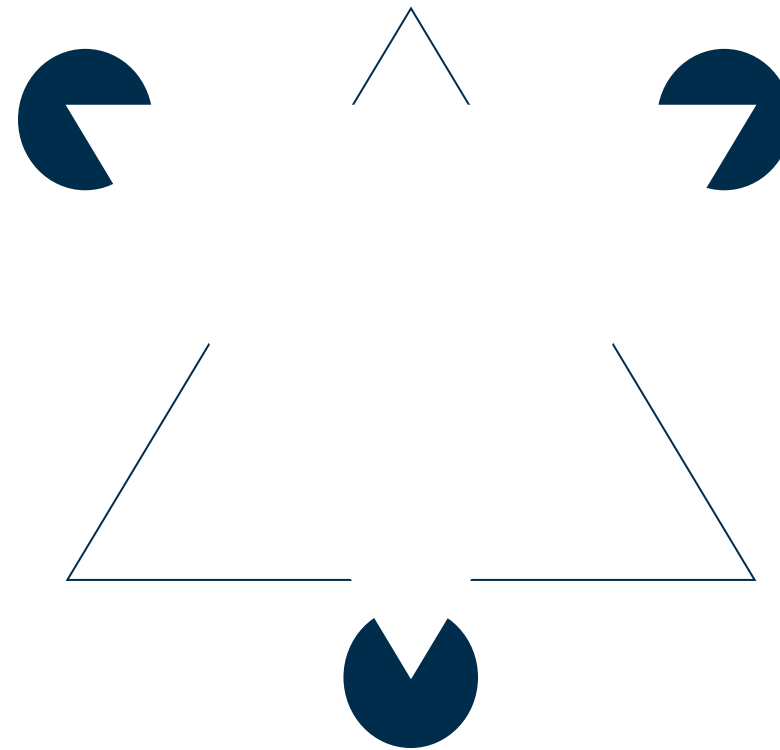
To make sense of it, you automatically separate it into known simple forms

□ simplification



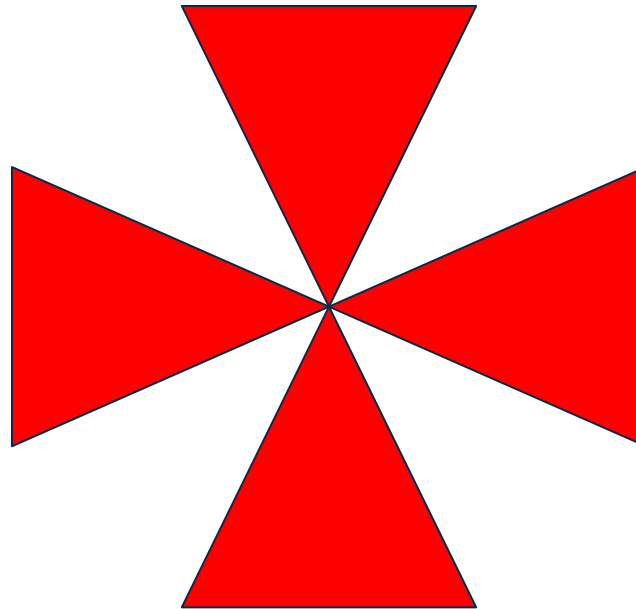
Gestalt Law – Closure I

Closure Principle – We tend to see things as complete objects even though there may be gaps in the shape of the objects
There seems to be a triangle inside...



Gestalt Law – Closure II

- **Closed figures** are perceived more easily than incomplete or open figures
- The figure either shows **four** triangles or **one** Maltese cross



Web Examples

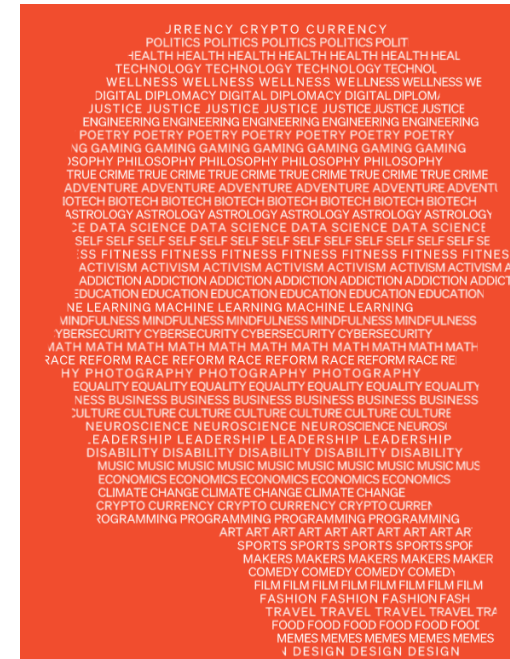
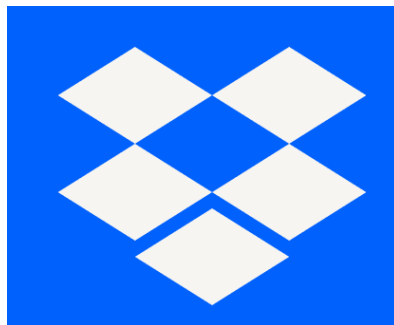
Medium Blog Page:

- Text make a picture
- Recognition: head
- Recall: text

Dropbox

- Five rough makes the logo

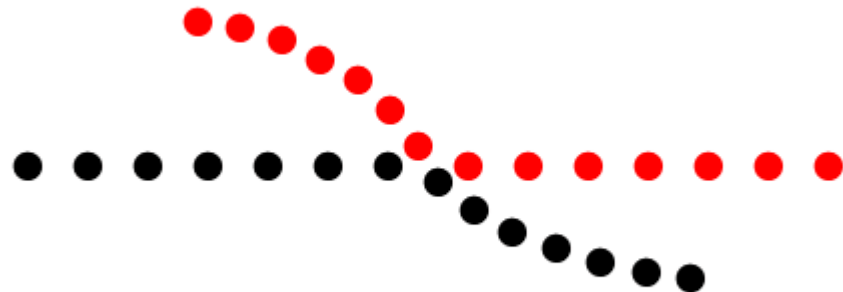
Source: Dropbox



Source: Medium.com

Gestalt Law – Continuity

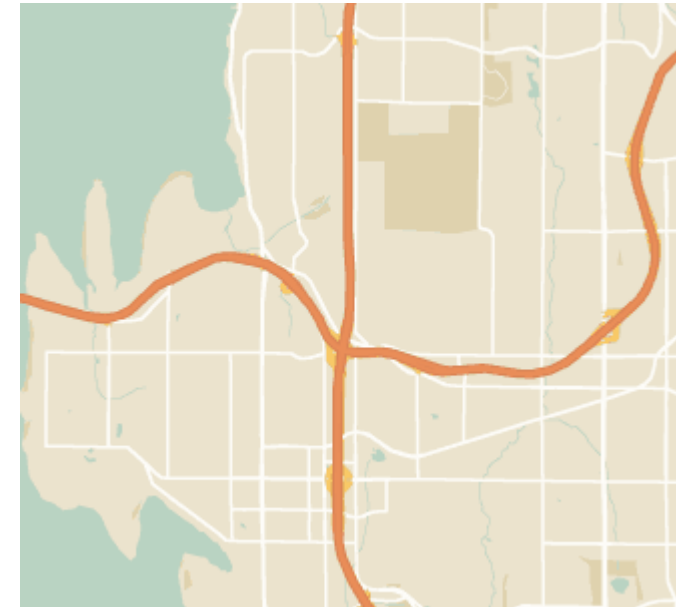
- We tend to perceive **smooth, continuous patterns** instead of disjoint, interrupted patterns.



→ red dots in the curved line seem to be more related to the black dots on the curved line than to the red dots on the straight horizontal line

[▶ image source](#)

Continuity Applied to UI Design



Gestalt Principles of Perceptions

- **Common Fate (Sub) Principle** – Objects that “move” together are seen as related

Unaligned Drop-Down Menus

The Principle of Common Fate ▼
The Principle of Common Fate ▼
The Principle of Common Fate ▼
The Principle ▼

Aligned Drop-Down Menus

The Principle of Common Fate ▼
The Principle of Common Fate ▼
The Principle of Common Fate ▼
The Principle ▼



[▶ image source](#)

Common Fate applied in UI Design

Menus are said
to be common fate



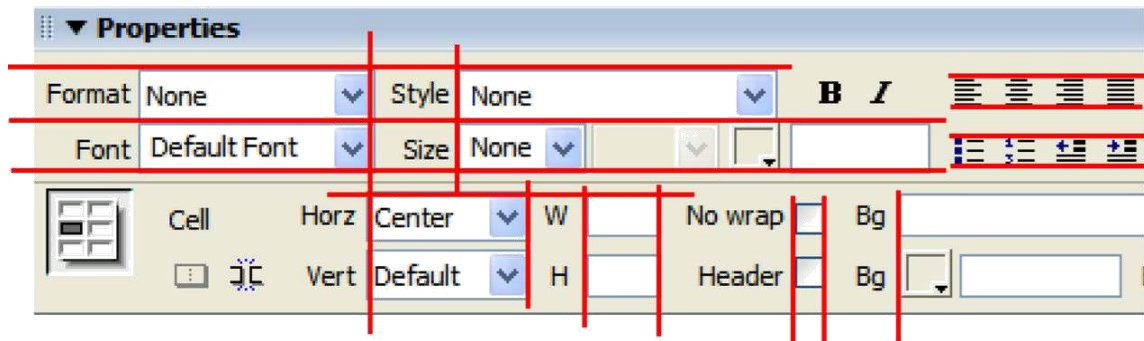
[image source](#)

Gestalt (Sub)Principles of Perceptions

Parallelism

Similar to Common Faith, without movement

Chunking by alignment, not by proximity

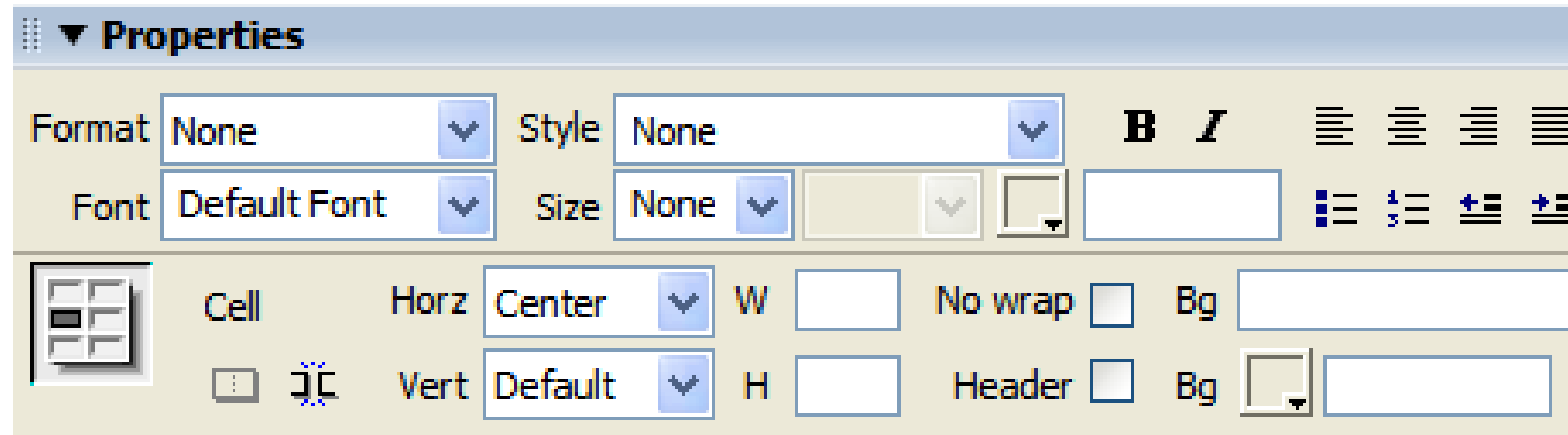


Gestalt Principles of Perceptions

You can overwrite the effect!

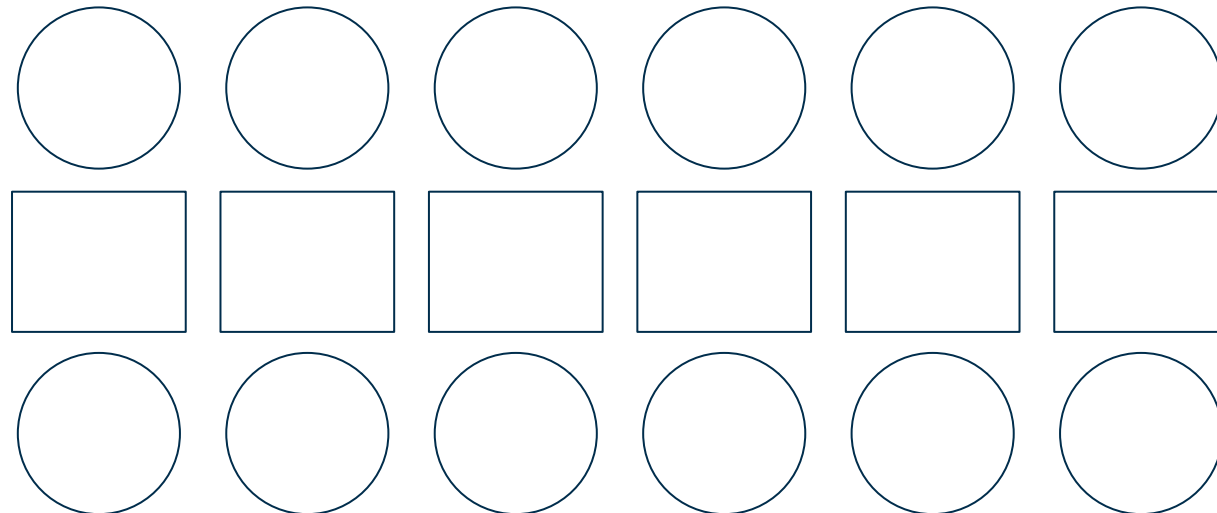
Property Pane from Macromedia's/Adobe's Dreamweaver

- Our eyes pick up all of the text boxes because of the **strong blue squares** and the **white areas** that they have in common
- ...which is bad design that overwrites the positive ordering / parallelism effect



Gestalt Law - Similarity

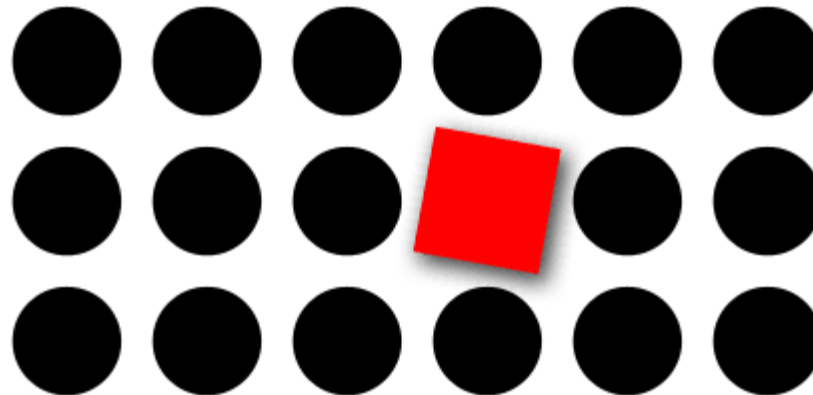
- Similar figures tend to be **grouped together**
- **Example:**
 - The figure shows two rows of circles and one row of rectangles
 - The figure above shows 6 columns of three-shape elements
 - (they are also parallel)



Gestalt Law - Similarity

– Other Example1:

- focal points
- quickly identify the unknown to alert us to potential danger



¹

▶ [source](#)

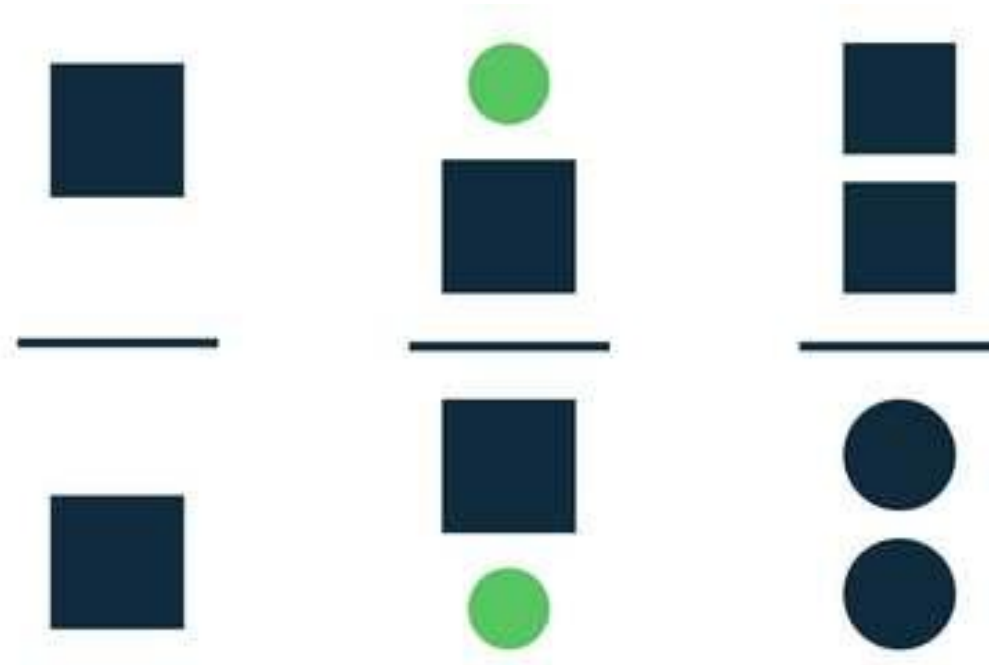
Similarity, Applications in UI Design

– What is similar here?



Gestalt Principles - Symmetry

Simplifies perception because only one form block needs to be perceived
The others are symmetric multiplications



[▶ image source](#)

Further Gestalt Law – Part-Whole Relationship



- **Same parts, different whole:** while both figures are made from the letter “H”, the whole is perceived differently:
 - ? The left figure is perceived as big H, while the right is not



Further Gestalt Principles of Perception

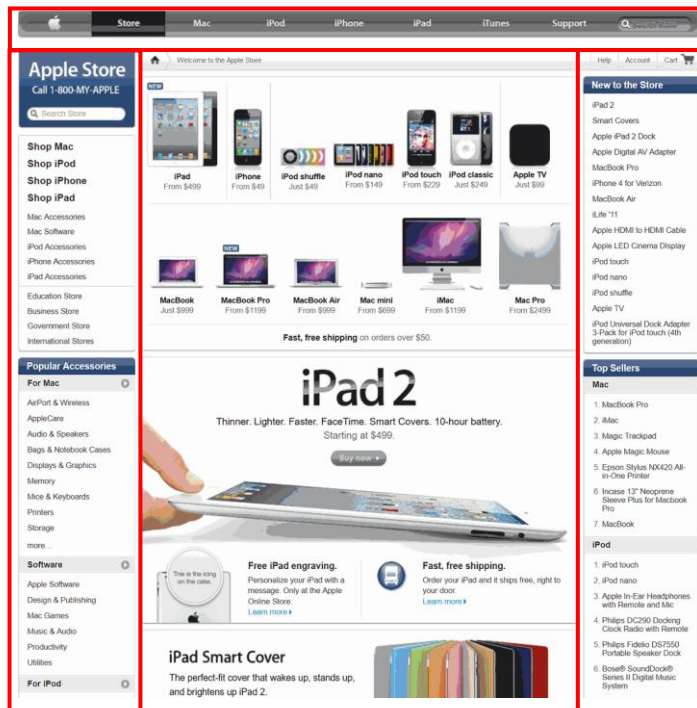
The Area Principle – Objects with small area tend to be seen as the figure, not the ground (also called the smallness principle)



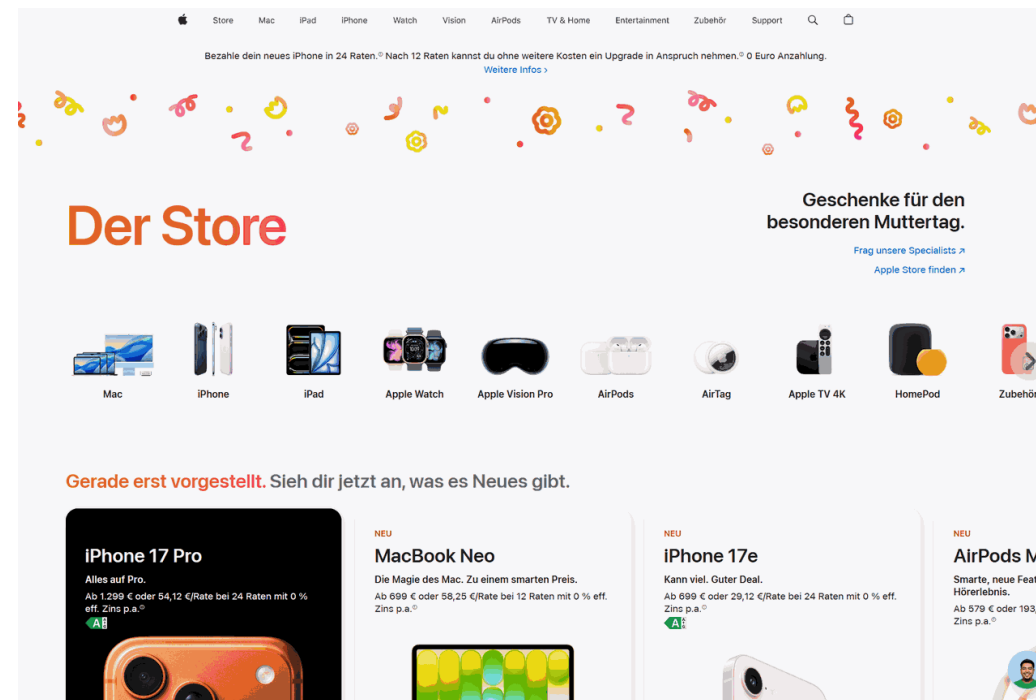
► [image source](#)

Further Gestalt Principles of Perception: Surroundedness

Surroundedness Principle – An Area that is surrounded will be seen as the figure and the area that surrounds will be seen as the ground (frames in red)



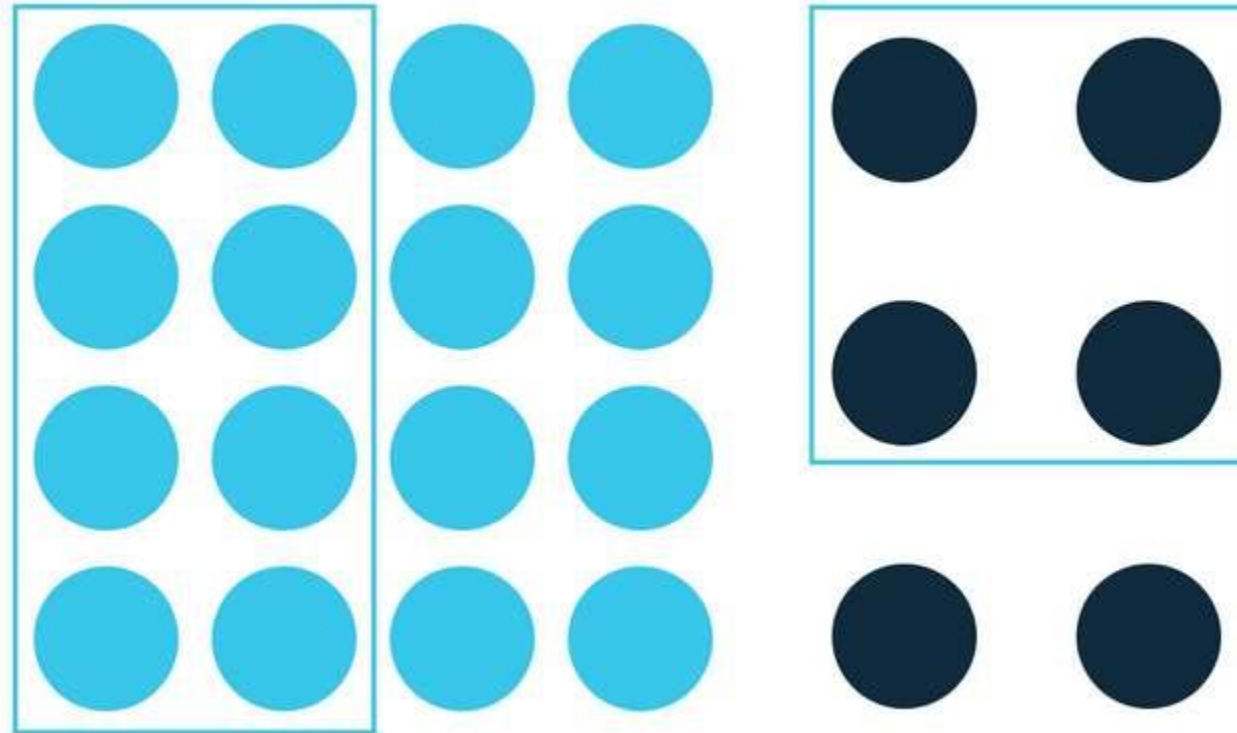
Apple Store 2008



Apple Store 2027

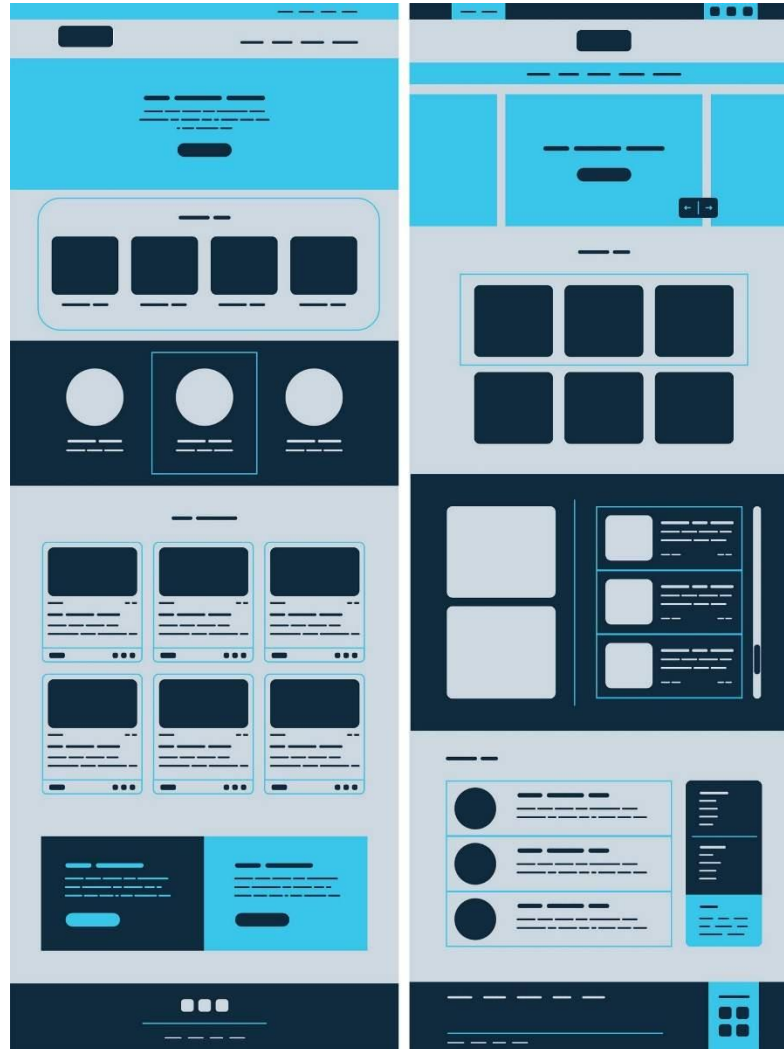
Gestalt Law: Common Region

Only by indication of regions
„Chunking“ by region
proximity



► [image source](#)

Common Region Principle in UI



Pingo

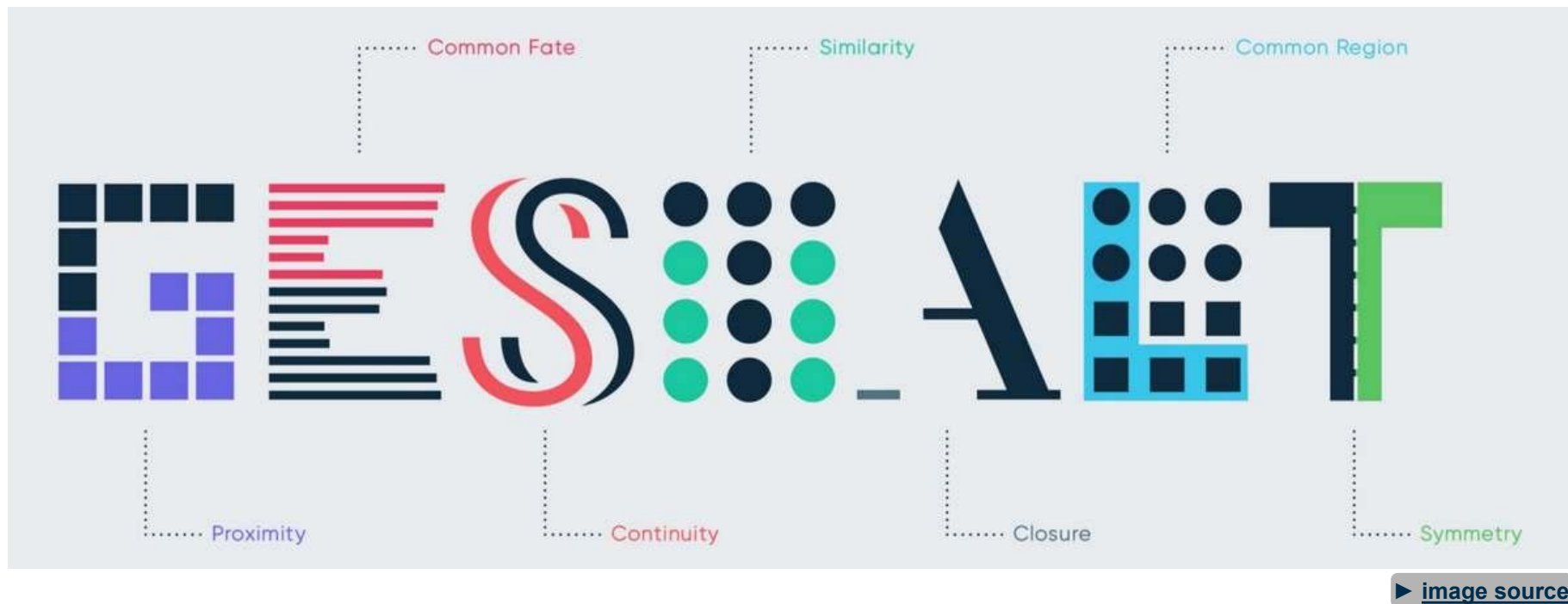
Question:

Answer question about “Gestalt”

Link: <https://pingo.coactum.de/589809>

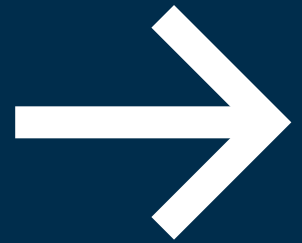


Gestalt Laws



Gestalt III

Principles of Perception



Other Principles of Perception – Stimulus Intensity



We respond first to the **intensity** of a stimulus and only then do we begin to process its **meaning**

1 3 9 7
4 8 6 2
5 7 1 3
7 9 3 1

Other Principles of Perception – Proportion



Proportion can be used to represent **logical hierarchies**

Heading Level 1

Heading Level 2

Heading Level 3

Heading Level 4

Heading Level 5

Heading Level 6

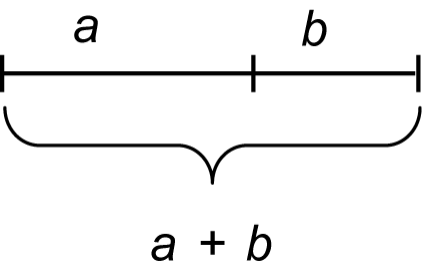
Other Principles of Perception – Aesthetics: Proportion / Golden Ratio

Usability is largely influenced by aesthetics (*)

Golden Ratio - The golden ratio expresses the relationship between two aspects of a form such as height to width and must equal 1.618 (=a/b)

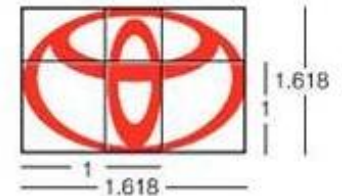
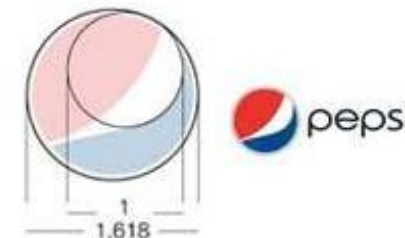
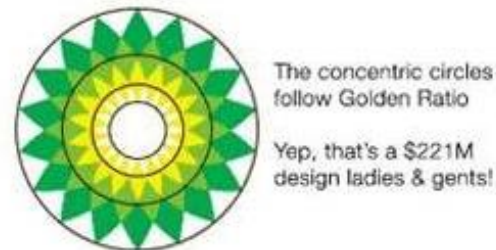
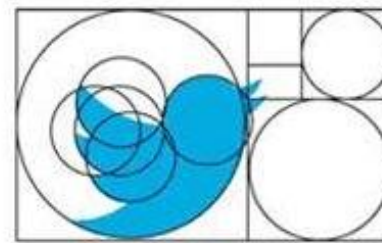
→ Seen as aesthetic

$a=0.618$
 $b=0.382$
 $\varphi=1.618$



$a + b$

$$\frac{a + b}{a} = \frac{a}{b} = \varphi$$



[▶ image source](#)

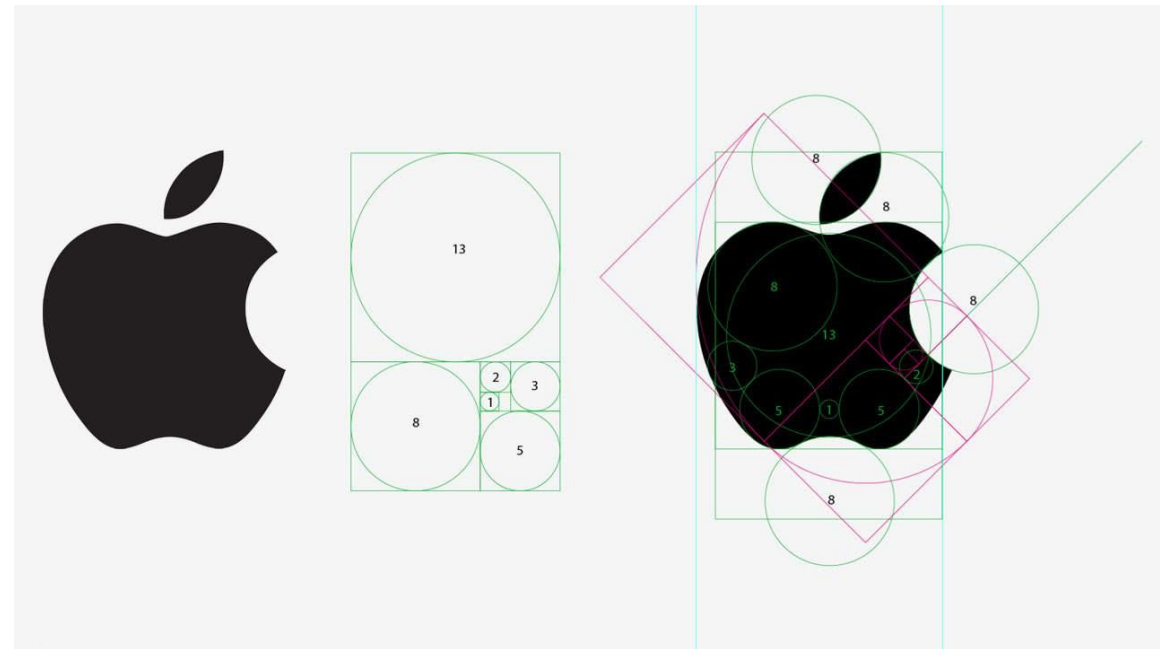
(*) Thüring, Manfred, and Sascha Mahlke. "Usability, aesthetics and emotions in human–technology interaction." *International journal of psychology* 42.4 (2007): 253-264.

Other Principles of Perception – Aesthetics in Practice

Example: Ipod and apple logo



► [image source](#)



► [image source](#)

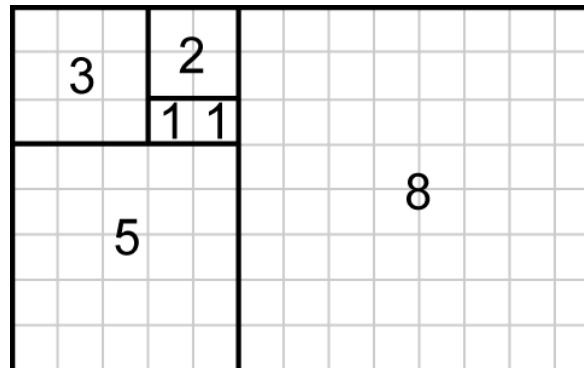


Other Principles of Perception – Aesthetics: Proportion Fibonacci

Fibonacci - A sequence of numbers in which each number is the sum of the two preceding numbers

The relationship between the numbers in the Fibonacci series is similar to the golden ratio
Seem as aesthetic

1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233, 377, 610, 987, ...



Other Principles of Perception – Screen Complexity



The measure of complexity developed by Tullis (1984) can be used to calculate the relative complexity, and therefore the difficulty, of a design

To calculate the measure of complexity for a particular (text list) screen, do the following:

- Place a **rectangle around every screen element**
- Count the number of elements and the number of **Columns** (vertical alignment points)
- Count the number of elements and the number of **Rows** (horizontal alignment points)
- **Layout Complexity = |R| + |C| - 2**

See common Gestalt principle

- Lowering above numbers reduces visual complexity
- But also think of other principles, e.g. Closure

Original formula is for pure text based input only

- But may be a good guideline for other type of displays as well

Other Principles of Perception – Screen Complexity



TEST RESULTS		SUMMARY:	GROUND
GROUND, FAULT T-G			
3 TERMINAL DC RESISTANCE			
>	3500.00 K OHMS T-R		
-	14,21 K OHMS T-G		
>	3500.00 K OHMS R-G		
3 TERMINAL DC VOLTAGE			
-	0.00 VOLTS T-G		
-	0.00 VOLTS R-G		
VALID AC SIGNATURE			
3 TERMINAL AC RESISTANCE			
-	8.82 K OHMS T-R		
-	14,71 K OHMS T-G		
-	62B.52 K OHMS R-G		
LONGITUDINAL BALANCE POOR			
-	39 DB		
COULD NOT COUNT RINGERS DUE TO			
LOW RESISTANCE			
VALID LINE CKT CONFIGURATION			
CAN DRAW AND BREAK DIAL TONE			

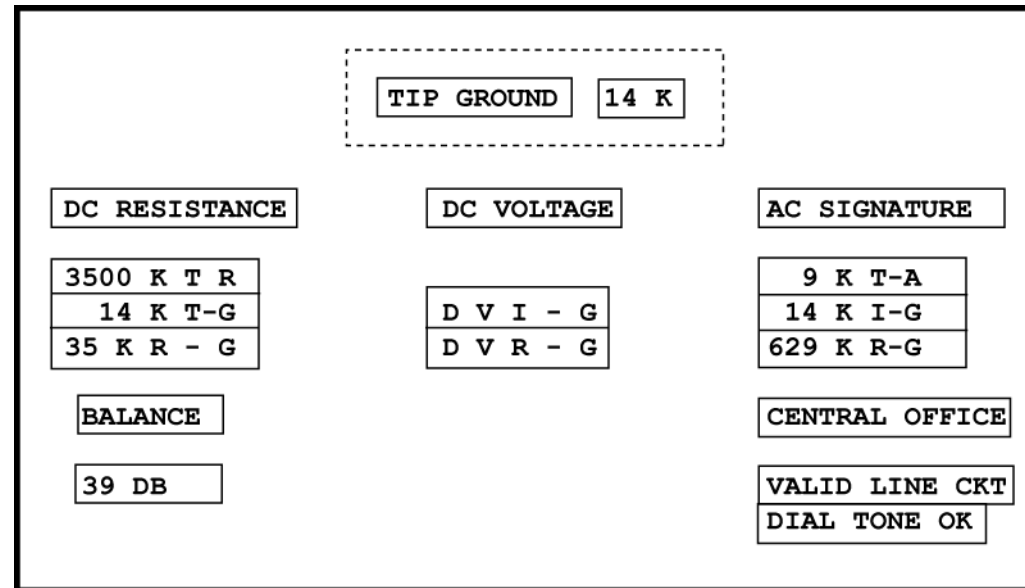
Horizontal
alignment
points

Vertical alignment points

TEST RESULTS		SUMMARY:	GROUND
GROUND, FAULT T-G			
3 TERMINAL DC RESISTANCE			
>	3500.00 K OHMS T-R		
-	14,21 K OHMS T-G		
>	3500.00 K OHMS R-G		
3 TERMINAL DC VOLTAGE			
-	0.00 VOLTS T-G		
-	0.00 VOLTS R-G		
VALID AC SIGNATURE			
3 TERMINAL AC RESISTANCE			
-	8.82 K OHMS T-R		
-	14,71 K OHMS T-G		
-	62B.52 K OHMS R-G		
LONGITUDINAL BALANCE POOR			
-	39 DB		
COULD NOT COUNT RINGERS DUE TO			
LOW RESISTANCE			
VALID LINE CKT CONFIGURATION			
CAN DRAW AND BREAK DIAL TONE			

Other Principles of Perception – Screen Complexity

Redesigned screen



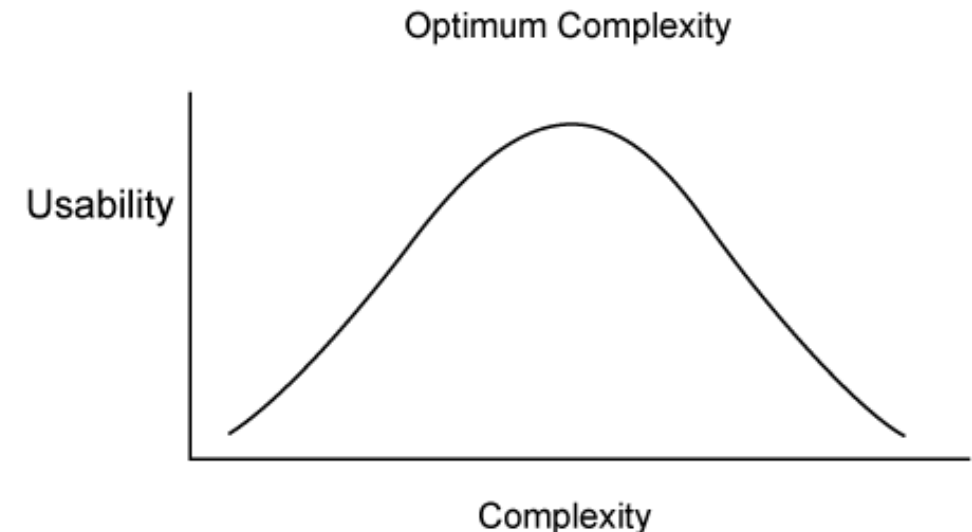
Other Principles of Perception – Screen Complexity

Complexity vs. Usability

Comber and Maltby (1997) found that both overly simple and overly complex screens were low in usability – which is in contrast to Tullis formula

Comber and Maltby defined usability in terms of the following three components:

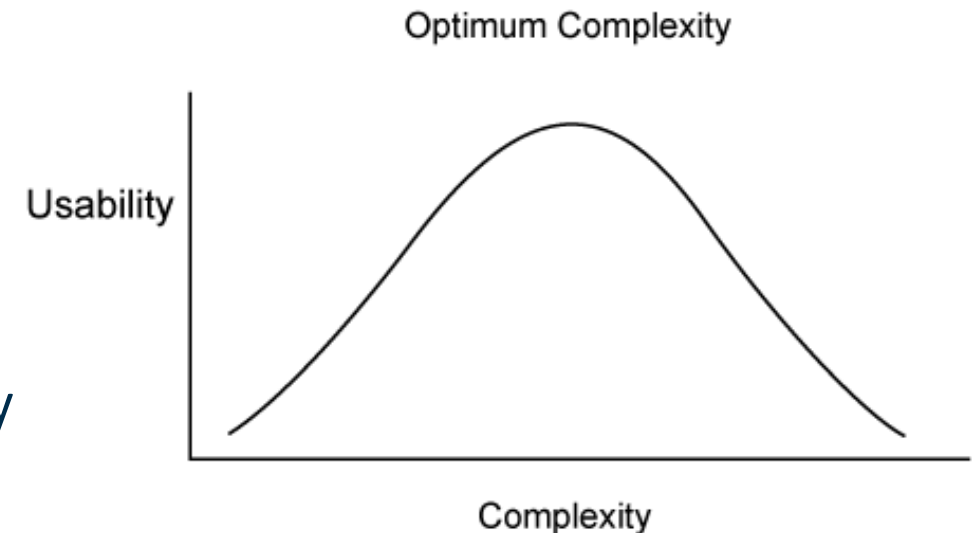
- Effectiveness
- Learnability
- Attitude



Other Principles of Perception – Screen Complexity

Comber and Maltby found **tradeoffs between usability and complexity**:

- As complexity decreased, predictability increased
- As complexity decreased, it became harder to differentiate among screen objects; the screen became artificially regular (→ app icons on handhelds)
- Decreased complexity meant that there were fewer ways to group objects
- Excessive complexity made screens look artificially irregular
- Increased complexity could occur from increased utility



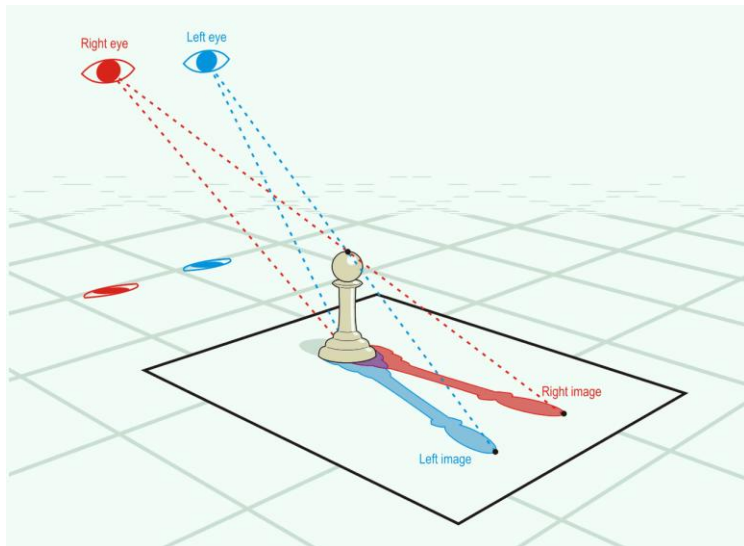
Gestalt IV

3D Perception



HIP: Depth Perception

- The arrangement of our visual sensors (eyes) allows us to perceive our world three-dimensionally.
- There two types of depth cues:
 - primary depth cues (relevant e.g. to immersive virtual reality systems)
 - secondary depth cues (more relevant e.g. to non-immersive applications such as games)



► [image source](#)

Designing Interactive Systems, D. Benyon, P. Turner and S. Turner, 2005, Addison-Wesley, p. 120

HIP: Depth Perception – Primary Depth Cues



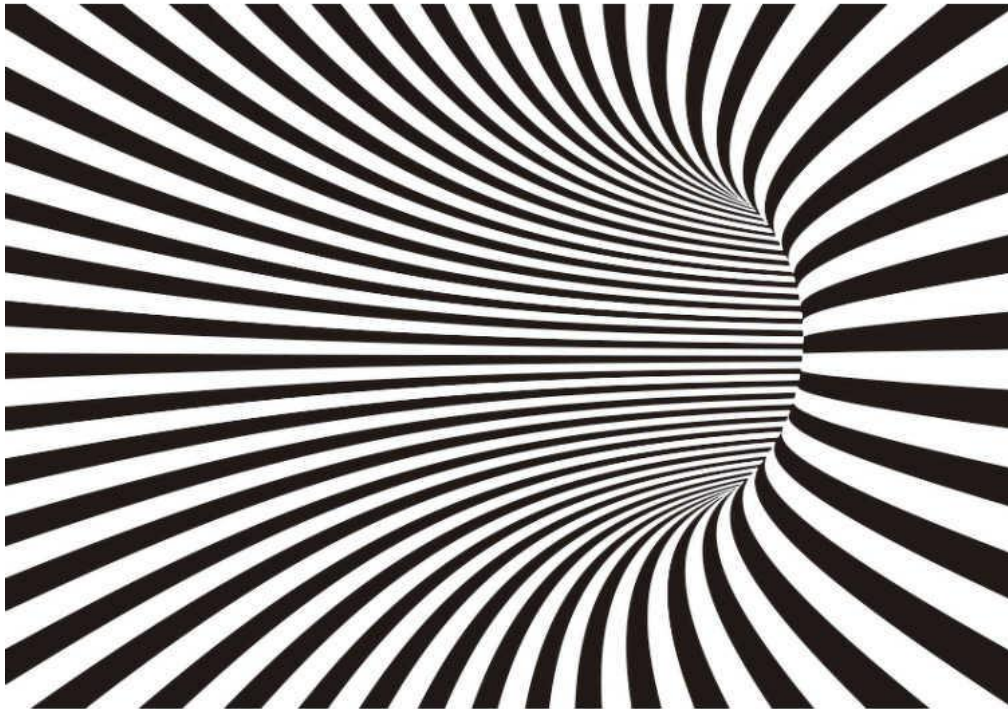
- The four primary depth cues are retinal disparity, stereopsis, accommodation and convergence:
- **retinal disparity:** As our eyes are approximately 7 cm apart, each retina receives a slightly different image of the world. This is processed by the brain and interpreted as distance information.
- **stereopsis:** Stereopsis is the process by which the different images of the world received by each eye are combined to produce a single three-dimensional experience.
- **accommodation:** This is a muscular process by which we change the shape of the lens in our eyes in order to create a sharply focused image. The information from the muscles is unconsciously used for depth information.
- **convergence:** Over distances of 2-7 meters we move our eyes more and more inwards to focus on an object at these distances. This process is used to provide additional depth information.

HIP: Depth Perception – Secondary Depth Cues

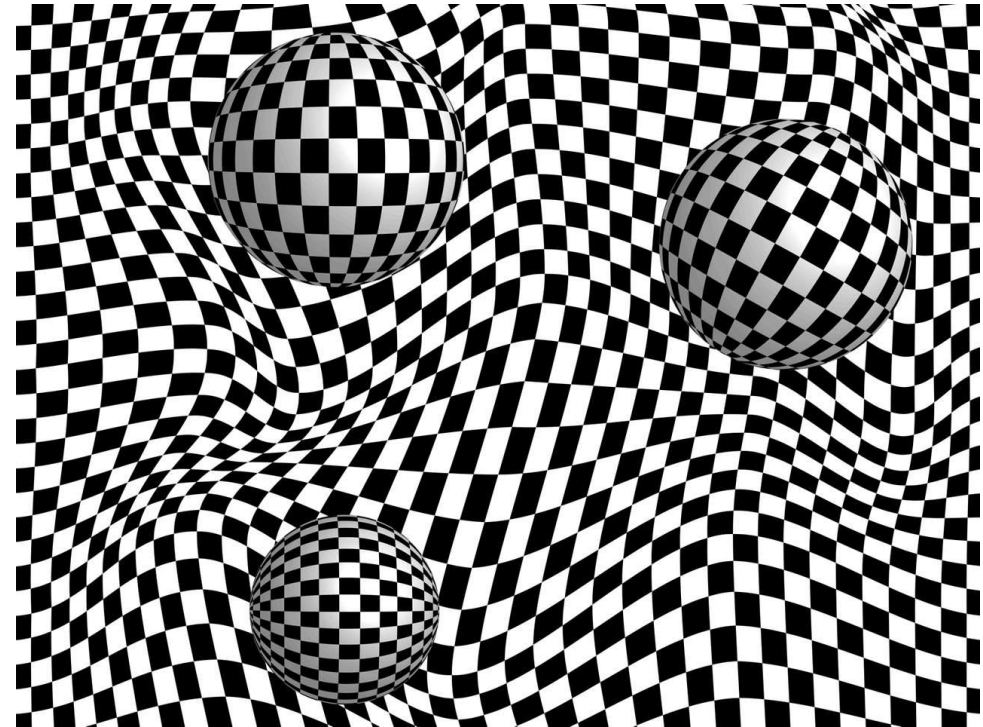
- The secondary depth cues are the basis for depth perception on two-dimensional displays. The secondary depth cues are:
- **light and shade**: an object with light and shadow improves the depth perception
- **linear perspective**: shadow on arrow
- **height in the horizontal plane**
- **motion parallax**: riding on a train and looking out of the window: near objects are perceived to flash by quickly, objects further away as slower
- **overlap**: e.g. overlapping windows in a GUI
- **relative size**: see sun + cloud
- **texture gradient**: textured surfaces appear closer



HIP: Depth Perception – Secondary Depth Cues

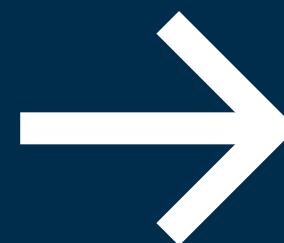


▶ [image source](#)



▶ [image source](#)

Visual Senses Colours



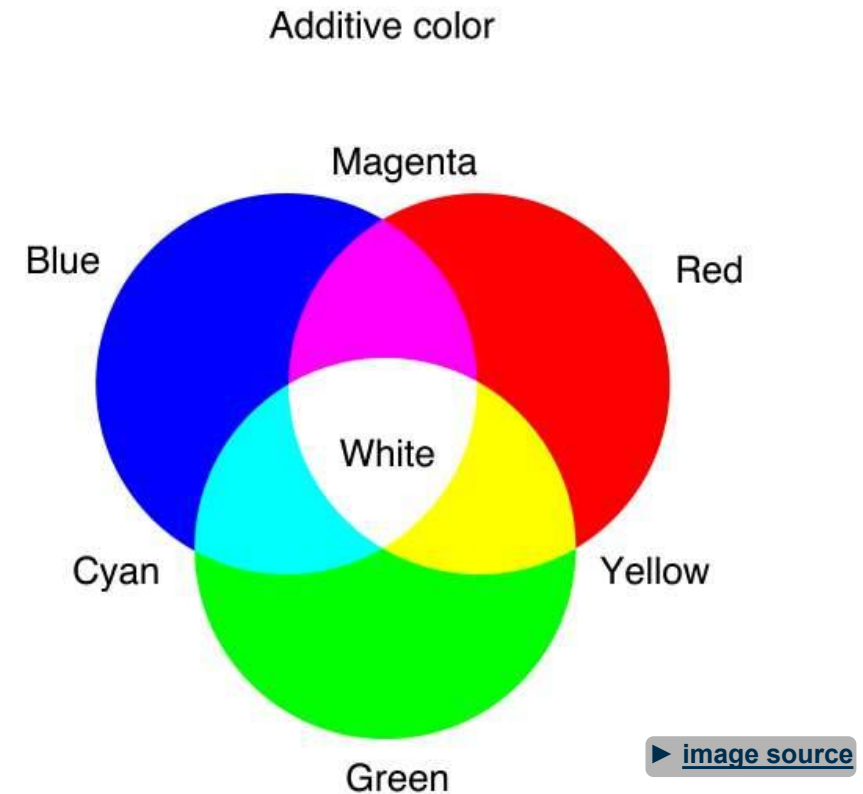
Colours: Additive

For mixing colours of light

Example: Displays

But: Information displayed with additive (active) light might be not as natural as reflected light

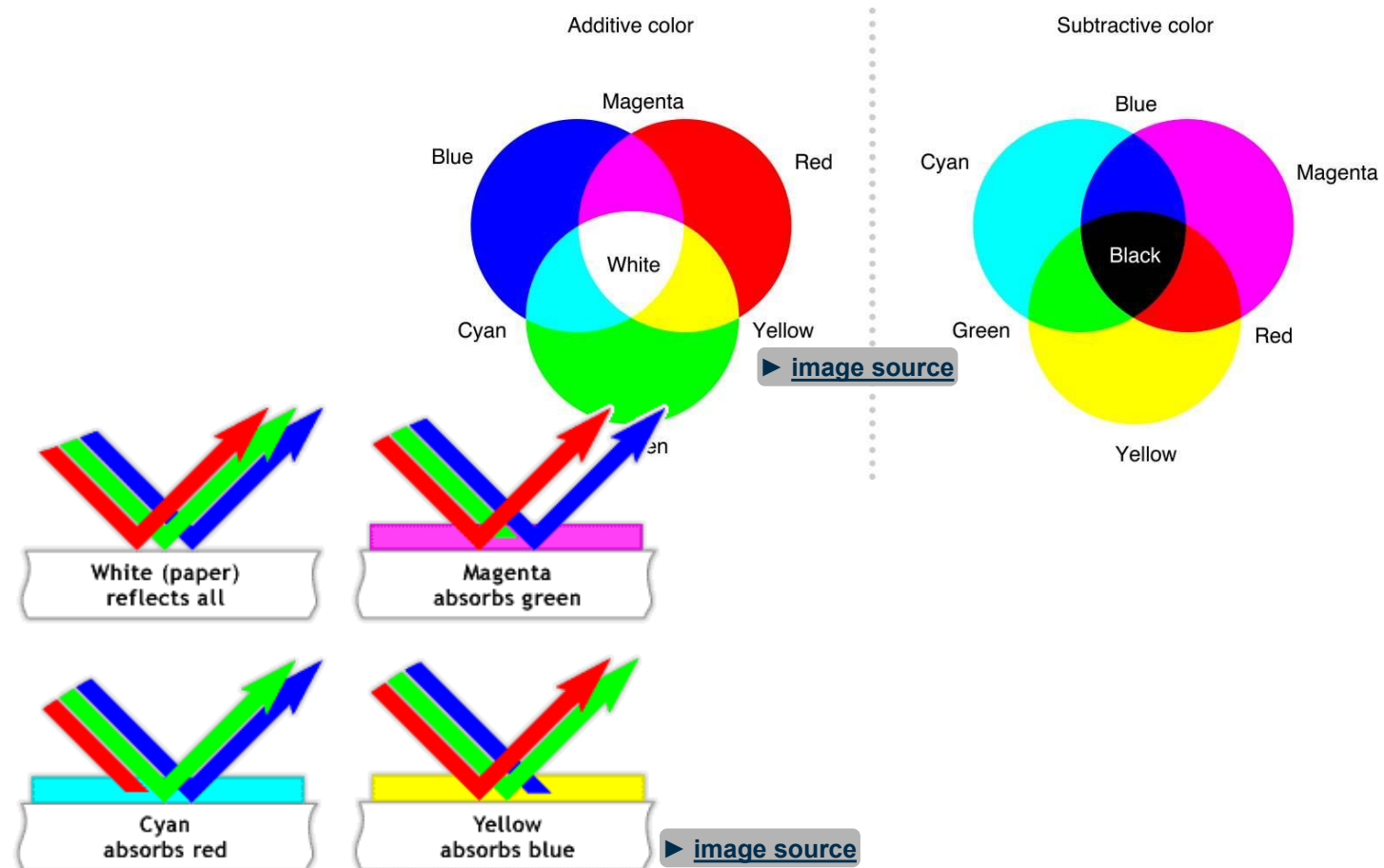
- A human is not used to look „into“ the light
- Natural perception is reflected light
- Additive coloured information presentation should be embedded into an environment that has equal brightness to avoid fatigue



Colours: Subtractive

For printed colours on material

- They reflect some of the light waves, while others are (more or less) absorbed
- Because you cannot hinder some reflection in all light spectra (that you like to absorb) and some filtering in your preferred spectrum, colours are not as shiny as in additive mode



Visual Limitations

- Blue Colour is on the edge of the spectrum (there is Ultraviolet, where you are least sensitive on it)
- Avoid using blue colour for small and tiny elements
- The ability to distinguish colour is directly related to the size of an object
- Shapes are identified by their edges. Edges are faster identified by lines than by colours
- Better colour perception in the middle of the focus
- About 8% of male (0.4% of female) have deficiencies to perceive colours correctly. Most deficiency is detection of green (www.vischeck.com)



► [image source](#)

Colours – attention and Emotion

Colours may cause problems leading to divided attention (as shown in test)

Colours may cause **emotional response**

- Emotional response varies from culture to culture.
- People from different age and sex have different preferences. E.g. children prefer colourful objects.
- Industries, professional communities, corporate have their own colour connotations



► [image source](#)



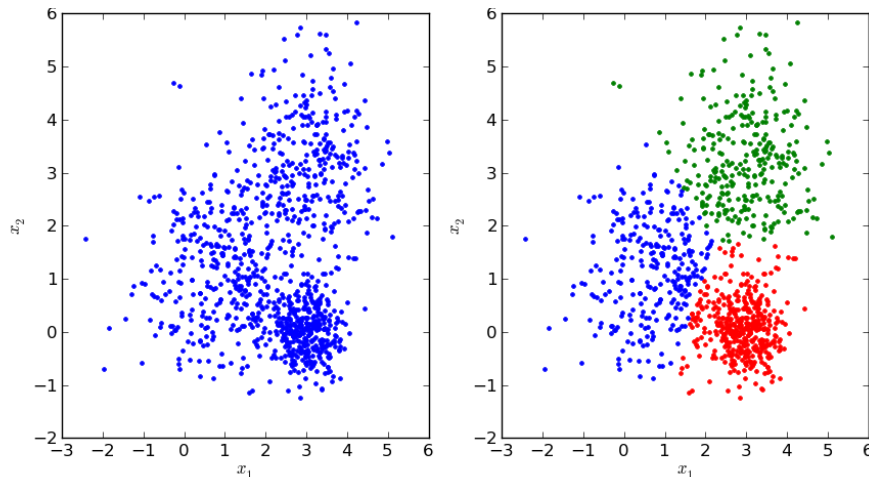
► [image source](#)



► [image source](#)

Colours for Design

For Clarification, Relation, Differentiation (e.g. Subway Maps)
Support finding (e.g. important element in a text)
Comprehension, Retention and Recall



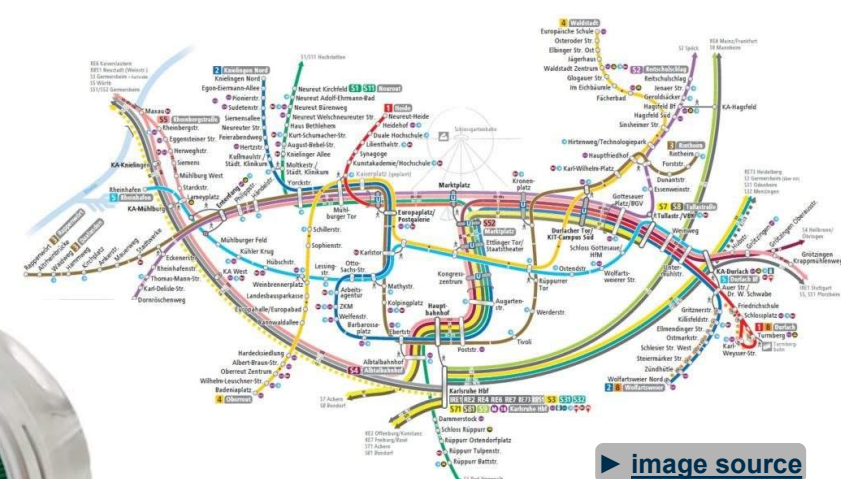
[image source](#)

**Impossible to separate
measurements if not colour coded**



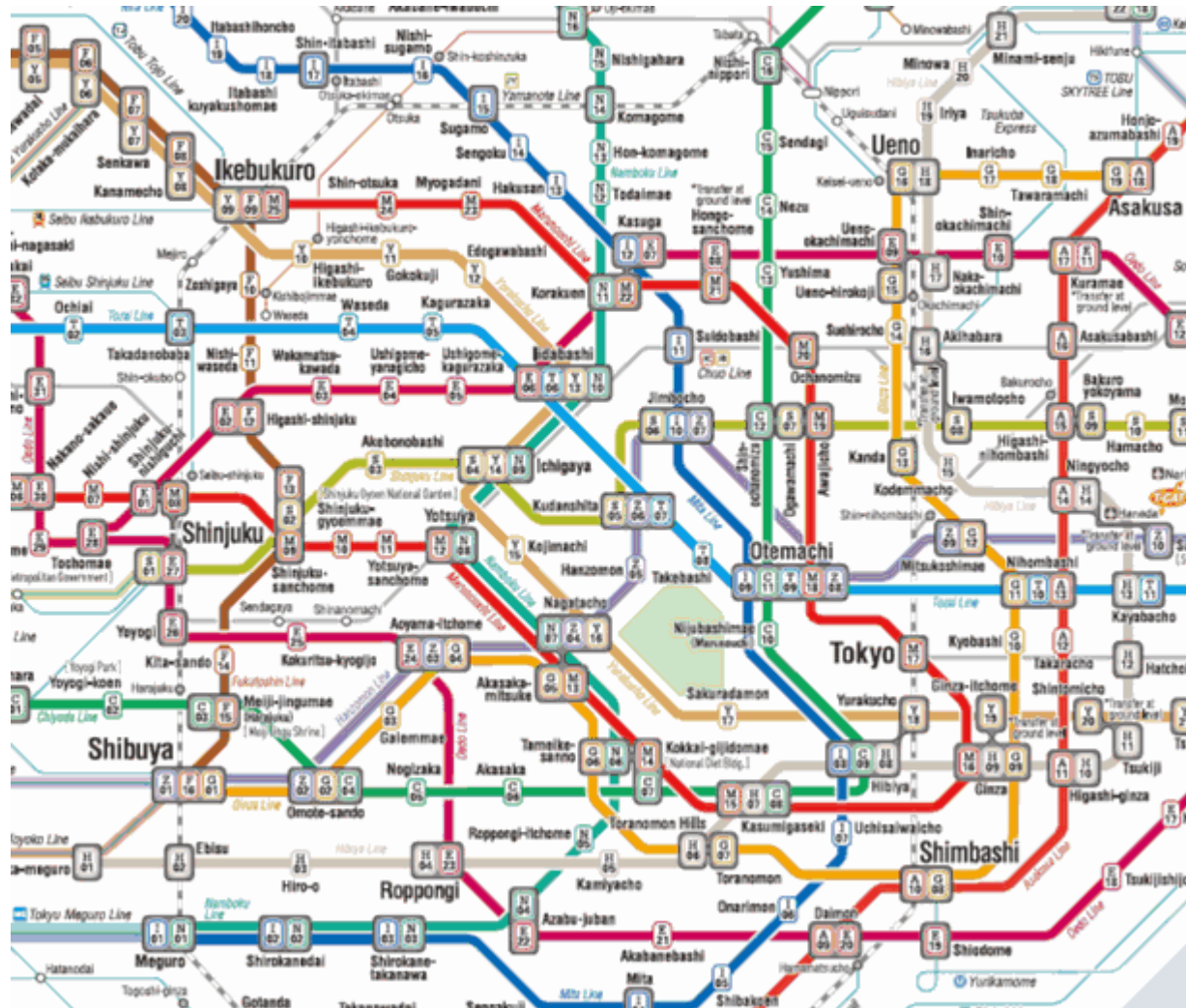
[image source](#)

**Difficult to remember what is a dangerous
level, but recall if colour coded**



[image source](#)

Tokyo Subway Map



Can improve recall, search-and-locate tasks, decision judgements → overall performance

Does not replace good structure

- A clear structure and presentation concept must already be present BEFORE colour is introduced

Differences to distinguish colour variations vary

- Between colours
- Between individuals

Targets may use highly diverse colours from those of their surrounding

- But this might be seen as distractive and not aesthetic
- The closer the colour areas are, the simpler we can separate
- The more similar a colour is, the more time it takes to separate
- We can only separate about 10 colours at one time

Western Colour conventions

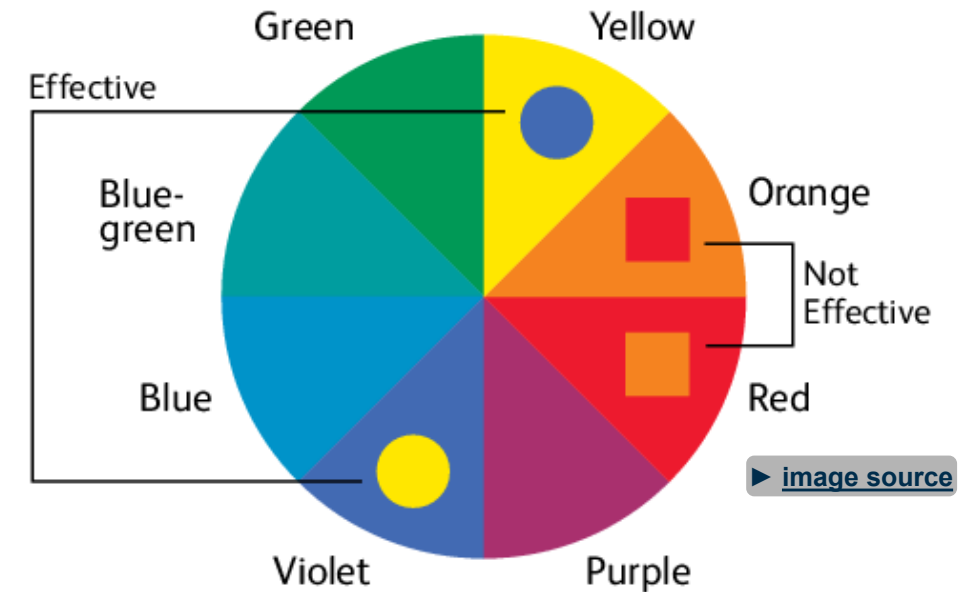
Red	Danger, hot, fire
Yellow	Caution, slow, test
Green	Go, okay, clear, vegetation, safety
Blue	Cold, water, calm, sky
Warm colours	Action, response required, proximity
Cool colours	Status, background information, distance
Greys, white and blue	Neutrality

Table 5.1 Some Western colour conventions.

Source: after Marcus, A. (1992) Graphic Design for Electronic Documents and User Interfaces, p. 84. © 1992 ACM, Inc. Reprinted by permission.

Optimal Colours

- Avoid any incompatible colour combination, e.g. bright red on blue
- Background to foreground contrast is important as well (see e.g. above)
- Good ideas, see figure
- Colour presentation and perception change with medium and display technology
 - Novel media may require rework
 - Most display media are not calibrated. What you see on your screen is NOT what others see
 - The light in the workspace may influence color perception as well. Be prepared for changes while the day.

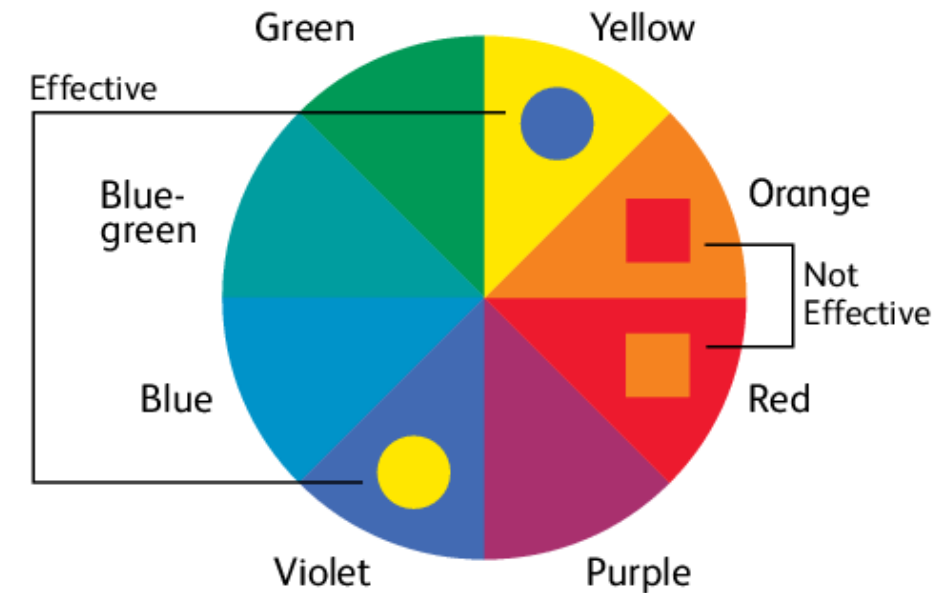


Optimal Colours

- Thorell & Smith found that red, blue, green and yellow are most beneficial in learning environments

How-to

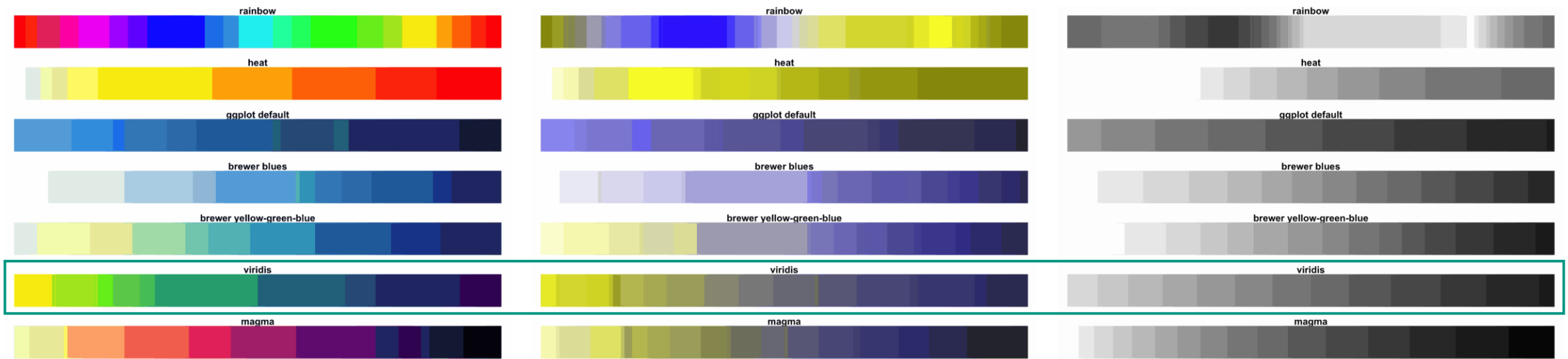
- First, use a colour code
 - E.g. max 4 main colours, each having max 4 variants
 - But Shneiderman suggested to use 4 colors on a single screen, and 5 max for one interface
 - This depends on the task: VERY Complex figures may require more colours, but average content should stay with 5
- Structure the content and identify the content that should be supported with a colour



► [image source](#)

Viridis Colour Map for colourblindness

Print well in grey scale and easier to read for colourblind people



[▶ image source](#)

Most used coding

- Hue: Basic Tone of colour
- Value/luminance/brightness: how „dark“ the colour appears
 - Like mixing black to the colour
 - Leads to „light“ or „dark“ colours
- Chroma/saturation: the grey-level of a colour
 - Like mixing grey to the colour
 - „muting“ the colour
 - Gray itself has 0% saturation (and hue 0)
 - Gray with 100% brightness is white, 0%=black
 - Human can perceive about 256 grey levels

Colour Models



There are many!

Computers today have mostly 24 or 32 bit colours spaces

Of interest now: Web Colour System

- Encoded like #00ff44 : 00 red, ff green, 44 blue, in hex
- Old systems and ePaper may be restricted to 256 colours, thus using a colour palette that may remap some of your colours!

Pingo

Question:

Answer question about “Colour”/”Farbe”

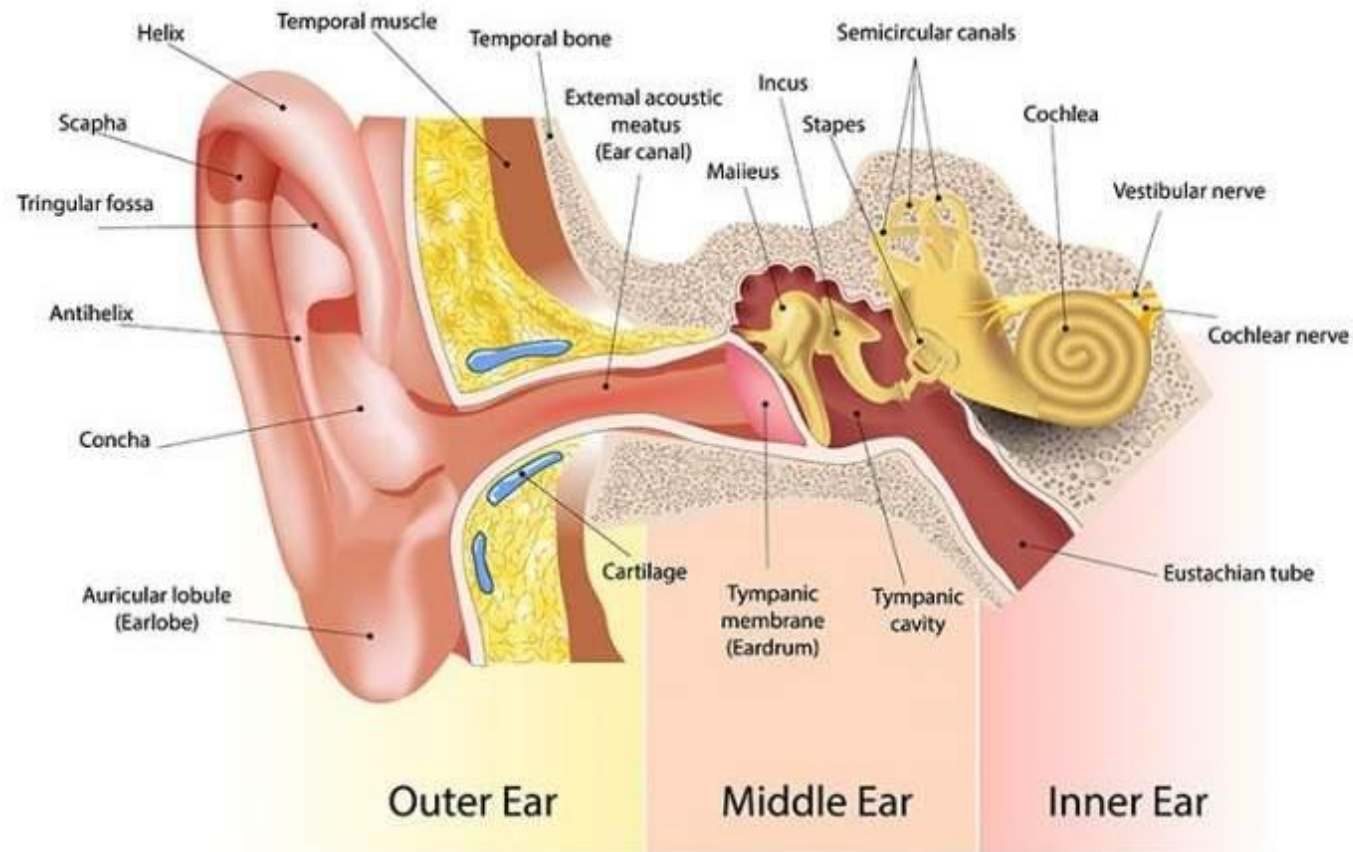
Link:

<https://pingo.coactum.de/589809>



Acoustic Senses →

Anatomy of the Ear



► [image source](#)

Loudness

Expressed in decibel (dB).

dB is a logarithmic (basis 10) scale

- E.g. 40dB is 10x louder than 30dB
- Whisper: 15dB
- Normal Conversation: 60-65 dB
- Car horn: 110 dB
- Rock concert: 120+ dB

Frequency range of hearing 20Hertz-20kHz

100dB Risk of Hearing Damage
90 Truck (50 meter)
80 Urban Background Noise
70 Highway (25 meter)
65 Speech
55 quite Urban
45 quite Rural
30 Whisper
20 Rustling Trees in Forest
10 Breathing
0 Threshold of Hearing

4 Stages

1. **Transduction** (Translation of sound vibrations into neural impulses)
2. **Auditory grouping**
 - Two steps
 - Segregation into separate streams
 - Integration of sound in coherent streams based on similar harmonic, frequency, location etc.
 - The first two processes may ignore some sounds → this is the psychoacoustic model that is used in lossy audio compression, e.g. mp3
3. **Scene Analysis**
4. **Interpretation**

Audio User Interfaces



Vision and Hearing go together

- But audio can give impression from area that are beyond vision

Audio can tell your eyes where to look

- But people become habituated to continuous sounds and ignore important sound based information
- Still: Continuous sounds requires resources from the human, even if it is a continuous sound, leading to additional stress

Audio can be perceived faster than visual cues

Audio is transitory! Vision is often not!

(Non-)Speech



Speech

We can speak faster than we can write

- But spoken interaction contains more redundancy in terms of content
- Speech requires knowledge of language

We can read faster than we can listen

- It is often therefore more efficient to read than to listen
- Listening is a linear task. Reading can jump!

Non-Speech

Auditory Feedback!

Problem:

- Beyond simple forms, sounds must be learned or familiar
- Can be annoying and ambiguous
- There are not so many expressions possible

Redundant Coding

- The „click“ sound when pressing a GUI button
- Can be used for emotion
 - The „miauw“ sound of my camera
- It is shown, that redundant coding aids recalling an interaction by giving additional association
- Can increase efficiency for tasks, as experts can react faster (on sound only)
- People with deficits (e.g. color blindness) may benefit from redundant coding
- Most important: Positive / Negative-Feedback

Speech applications

- We won't go into details in this lecture

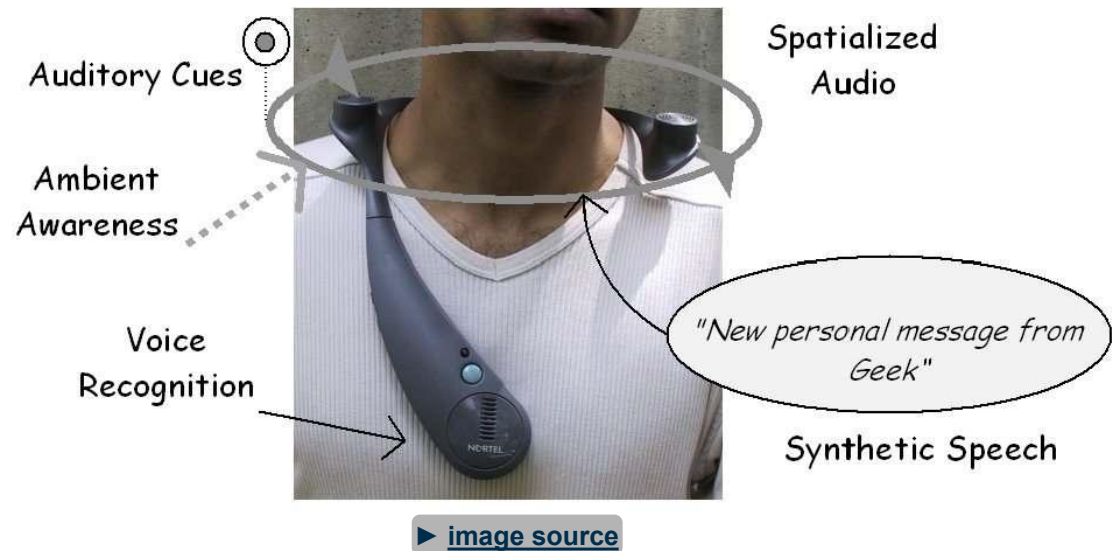
Earcon

- Based on musical sounds

Audio Wearable and Earcons

Nomadic Radio (1995-2000)

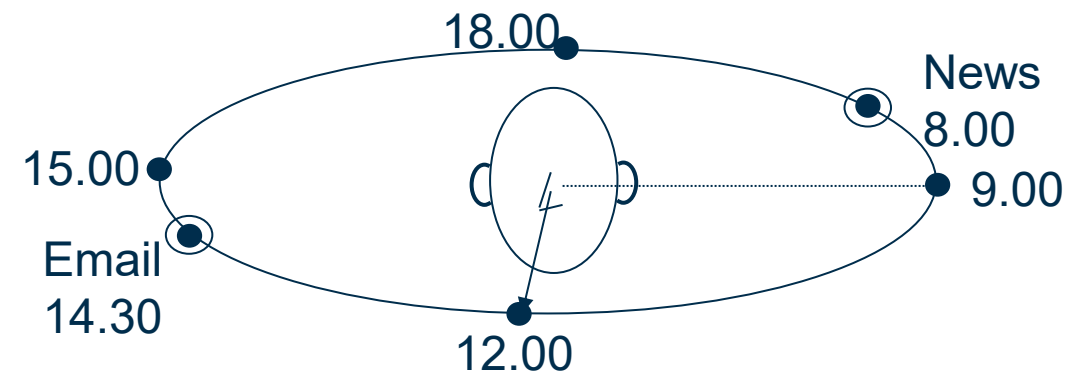
- Sawhney & Schmandt, MIT MediaLab
- Audio-only Wearable for ubiquitous communication
- output: Peripheral noise (Cues) and synthetic speech (messages, answers to commands)
- Input: Speech-recognition
- directed audio with speakers near humans head for 360° output



Nomadic Radio: Navigation

Active Interaction

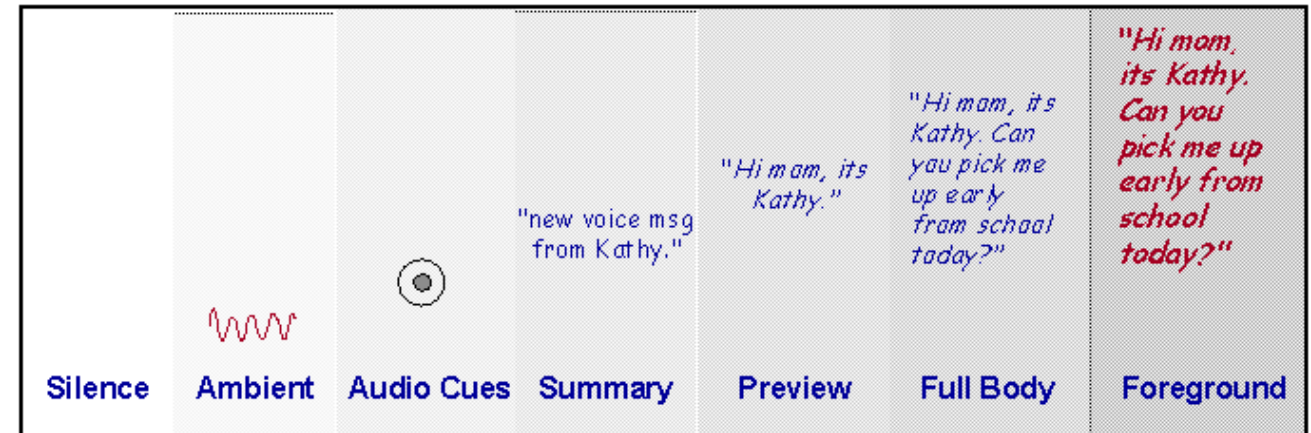
- Vocabulary for speech navigation
- Messages are **ordered in space** using 3D-Audio
- **Body-stabilized**, e.g. based on reception time of message
- Simultaneous audio: e.g. email in foreground, news in the background



Nomadic Radio: Audio Cues and Messages

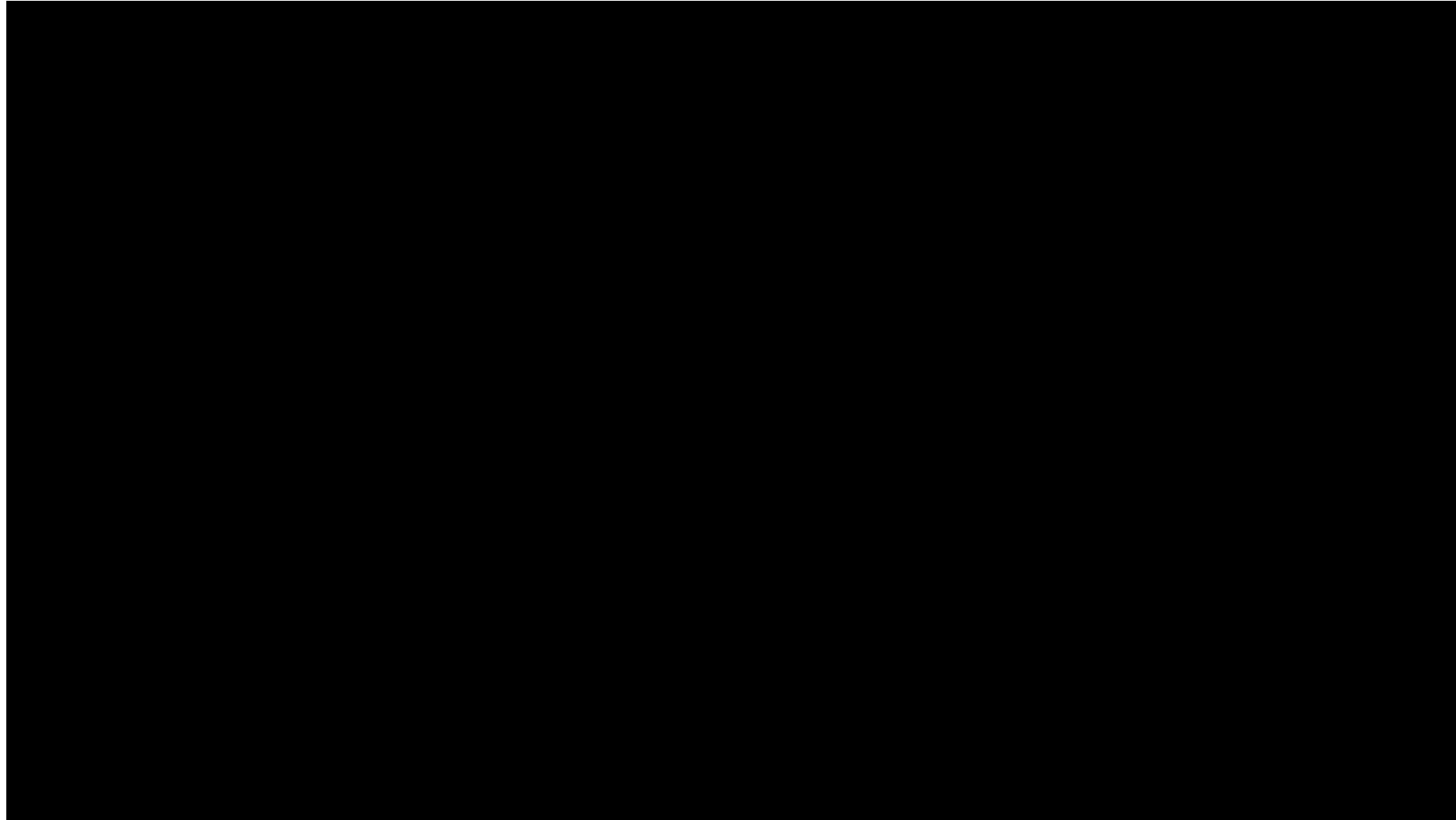
Passive Interaction, peripheral recognition

- Scalability
- Enough information without disruption
- **Seamless** refinement of information through blending



► [image source](#)

Top 100 most clicked ACM Videos



Pingo

Question:

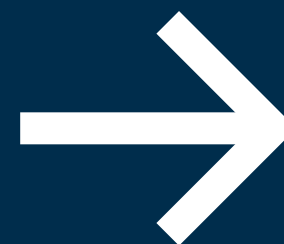
Answer question about “Audio”

Link:

<https://pingo.coactum.de/589809>



Touch



Why haptics in wearable computing?

HOW

Examples

- Compact actuators at the wrist, hand, arm, or torso
- Cues encoded by rhythm, intensity, timing, and location
- Modalities: vibration, stretch, pressure, thermal, electrical stimulation

WHAT IT ENABLES

e.g.

- Notifications and confirmations
- Navigation, warnings, and situational awareness
- Skill feedback, rehabilitation, and sensory substitution

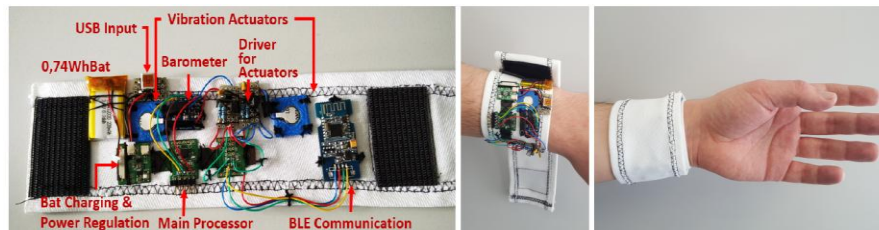
Examples: from consumer products to research prototypes



Industry: smartwatch haptics

Taps for alerts, confirmations, and silent interaction.

Source: Apple Watch Support



Publication: RüttelFlug

Wrist-worn tactile variometer: vertical velocity is sensed by pressure and displayed by two vibration actuators.

Source: Pescara, Beigl & Budde, ISWC 2016, Fig. 2



Publication: VTS Glove

Vibrotactile stimulation for upper-extremity rehabilitation after stroke.

Source: Seim et al., JNER 2021

Rececpotrs are specialized biological filters

Receptor subtype	Hair follicles	Meissner corpuscle	Pacinian corpuscle	Merkel cell-neurite complex	Ruffini corpuscle	C-fibre LTM	Mechano-nociceptor Polymodal nociceptor
Skin stimulus 							
Afferent response 	RA, LT	RA, LT	RA, LT	SA, LT	SA, LT	SA, LT	SA, HT
Stimulus 							
Receptive field 							
Perceptual functions Skin movement Skin motion; detecting slipping objects Vibratory cues transmitted by body contact when grasping an object Fine tactile discrimination; form and texture perception Skin stretch; direction of object motion, hand shape and finger position Pleasant contact; social interaction Skin injury; pain	Skin movement	Skin motion; detecting slipping objects	Vibratory cues transmitted by body contact when grasping an object	Fine tactile discrimination; form and texture perception	Skin stretch; direction of object motion, hand shape and finger position	Pleasant contact; social interaction	Skin injury; pain

In a nutshell

- stimulus class, afferent response are different, among receptor cell types,
- based on receptor anatomy and afferent type,
- leading to different receptive fields and perceptual functions.

Key idea

A haptic signal never acts directly on “touch”; it is first filtered by skin mechanics and receptor properties.

* RA=Rapidly Adapting, SA=Slow Adapting, LT= Low Threshold, HT=High Threshold
Red circles: cell types (e.g. C-fibre)

Receptor capabilities derived from

Each receptor type preferentially responds to different mechanical events and generates different perceptual qualities.

RECEPTOR	PREFERRED STIMULUS	PERCETUAL EFFECT
Hair follicle	Light touch	Skin motion
Meissner	Dynamic deformation	Flutter, slip, moving skin
Pacinian	Vibration	High-frequency vibration, tool texture
Merkel	Pressure / indentation	Form, edges, fine spatial discrimination
Ruffini	Stretch	Skin stretch, hand shape, movement direction
C-fibre LTM	Gentle touch	Pleasant social contact
Nociceptor	Damaging forces	Pain / warning signal

→ Select an actuator by the receptor profile it should reliably engage—not by component class alone.

Deriving effects from receptor profiles

RA / LT

Onset, flutter, slip; short pulses, ramps, and vibration bursts

SA / LT

Pressure, form, stretch; static indentation, shear, continuous stretch

SA / HT

Pain / warning; a safety boundary, not a target channel

Receptor subtype	Hair follicles	Meissner corpuscle	Pacinian corpuscle	Merkel cell-neurite complex	Ruffini corpuscle	C-fibre LTM	Mechano-nociceptor Polymodal nociceptor
Afferent response	RA, LT	RA, LT	RA, LT	SA, LT	SA, LT	SA, LT	SA, HT
Stimulus							
Receptive field							
Perceptual functions	Skin movement	Skin motion; detecting slipping objects	Vibratory cues transmitted by body contact when grasping an object	Fine tactile discrimination; form and texture perception	Skin stretch; direction of object motion, hand shape and finger position	Pleasant contact; social interaction	Skin injury; pain

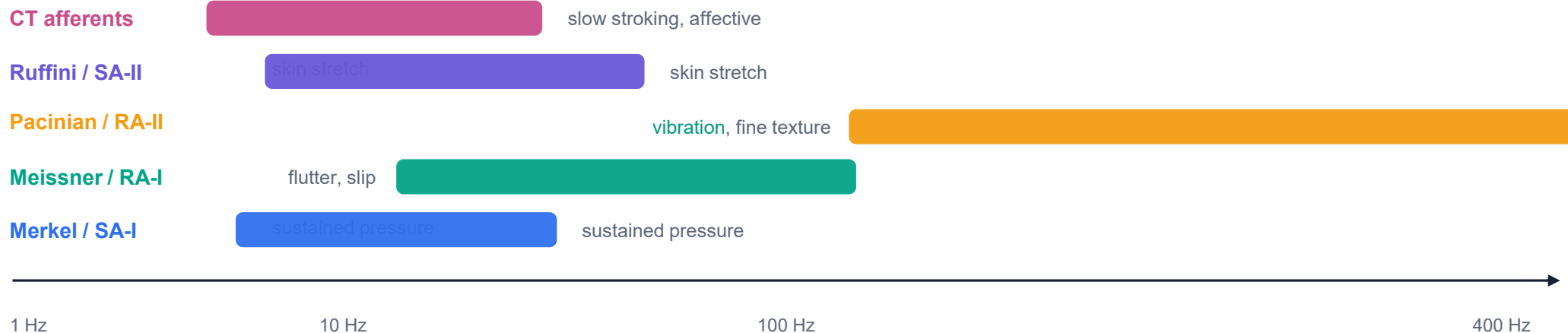
→ Define the desired perception first, then choose stimulus form, body site, and actuator.

Delmas, Patrick, Jizhe Hao, and Lise Rodat-Despoix. "Molecular mechanisms of mechanotransduction in mammalian sensory neurons." *Nature Reviews Neuroscience* 12.3 (2011): 139-153.

Receptors vs. Speed of mechanical sensation vs. use



- Specific Cell types are better for specific use
- They also cover different speed of sensations



Two types of touch: Discriminative vs. affective touch

Discriminative touch

- Object identity, location, edges, texture, slip
- Mostly fast, **A β** afferents
- Design goal: reliable, discriminable information

Affective touch

- Pleasantness, social meaning, comfort
- **C tactile** afferents: slow, especially relevant on hairy skin
- Design goal: account for context, body site, and speed

Not every strong stimulus is informative; not every weak stimulus is pleasant.

Actuator choice follows the desired receptor effect

How to select the right actuator for a required haptic interface?

- Technologies are means, not the starting point for design.
- Rule: use case → target perception → receptor profile → body site → actuator + signal.

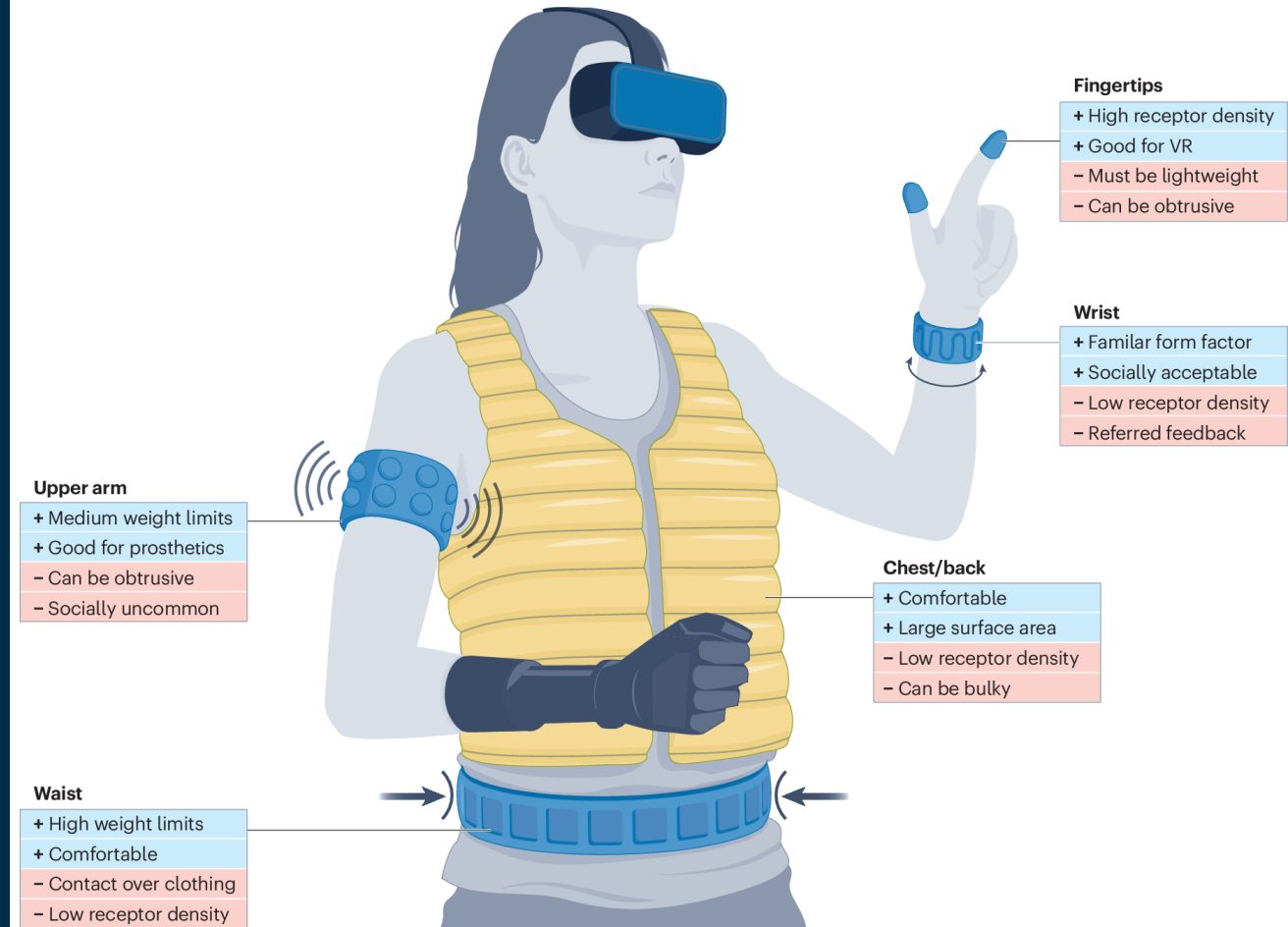
MODALITY	EXAMPLE TECH	DESIGN and AFFECT
Vibration	Eccentric Rotating Mass, Linear Resonant Actuator, Piezo	Pacinian / RA: salient, compact
Stretch / shear	tendon, Shape Memory Alloy, electromagnetic	Ruffini / proprioceptive cueing
Pressure / shape	pneumatic, jamming	Merkel / SA: form and pressure
Friction	electrostatic, ultrasonic	Texture and slip cues at surfaces
Thermal	Peltier, heater	Material realism; slow and safety-critical
Electrical	electrotactile, EMS	thin, calibrated, comfort-sensitive

Wearability

- Wearables must be ergonomic
- Wearables must be socially accepted
- Signal must be „perceivable“
- Studies have looked into this →

Fleck, J. J., Zook, Z. A., Clark, J. P., Preston, D. J., Lipomi, D. J., Pacchierotti, C., & O'Malley, M. K. (2025).

Wearable multi-sensory haptic devices. *Nature Reviews Bioengineering*, 3(4), 288-302.



Distribution of Mechanoreceptors

- Mechanoreceptors are not equally distributed among the body
- There is little complete knowledge
- The figure shows a distribution of some receptor classes only

- Taube Navaraj, W., García Núñez, C., Shakthivel, D., Vinciguerra, V., Labeau, F., Gregory, D. H., & Dahiya, R. (2017). Nanowire FET based neural element for robotic tactile sensing skin. *Frontiers in neuroscience*, 11, 501.

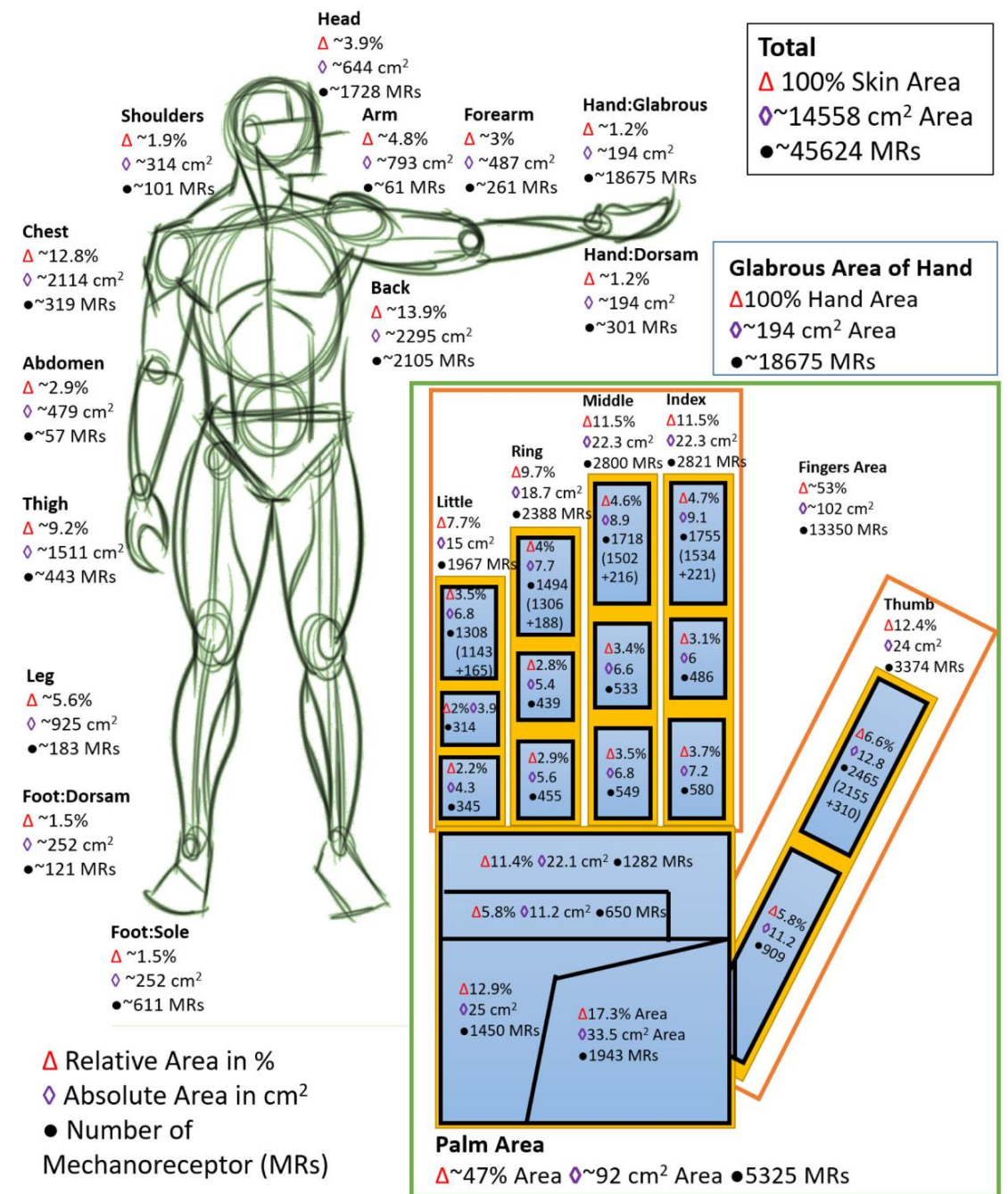
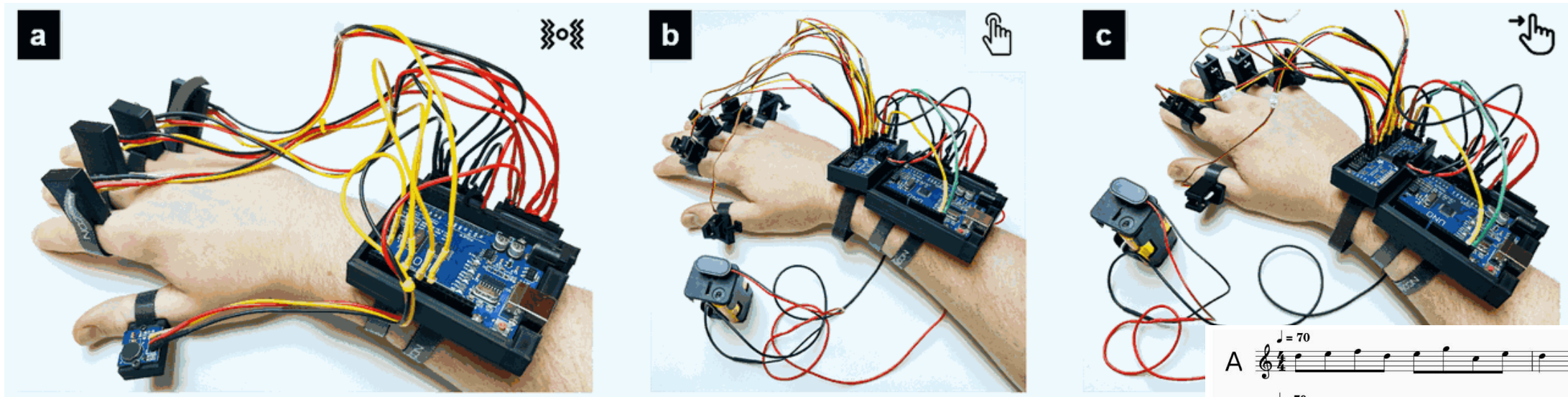


FIGURE 1 | The distribution of mechanoreceptors in various parts of human body. (Inset: 100% glabrous area of hand, which corresponds to 1.2% of the total skin area). The number of mechanoreceptors indicate the typical number of sensors required in various parts to achieve a full body touch/pressure sensing to mimic human body.

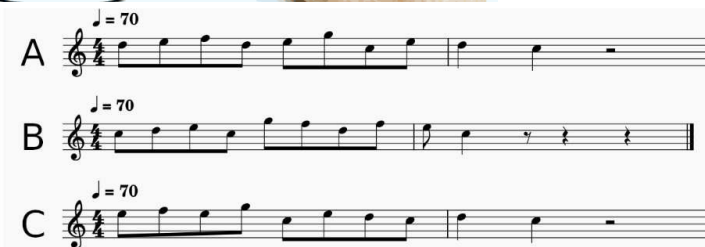
Best sensation to learn piano via tactile sensation passively (Gweld game)?

Various Sensations: vibration (a), tapping (b), and stroking (c)

Various User Groups: novice to expert



Fang, L., Müller, T., Pescara, E., Fischer, N., Huang, Y., & Beigl, M. (2023). Investigating passive haptic learning of piano songs using three tactile sensations of vibration, stroking and tapping. *Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies*, 7(3), 1-19.



Results

Effectiveness and Error Rate

Same Error rate for discriminative and affective sensations

- *ANOVA results cannot reject hypothesis*
- Likely to be the same ($F \ll 1$)

Error Rate differences in detail

- Indication that “pure” haptic sensation (affective OR discriminative) might perform better than “mixed” ones like vibration

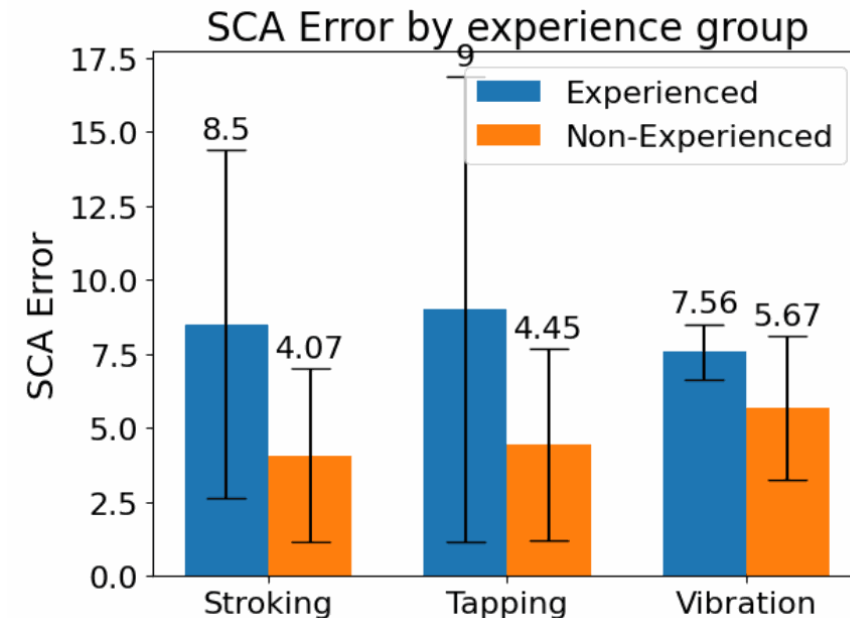
	F	<i>p</i>
SCA	0.55	0.58

Sensations	Mean	St. Dev.	Max	Min
Stroking	4.6	3.7	13.3	0
Tapping	5.1	4.4	20.0	0
Vibration	5.9	2.5	11.3	0

Further Results

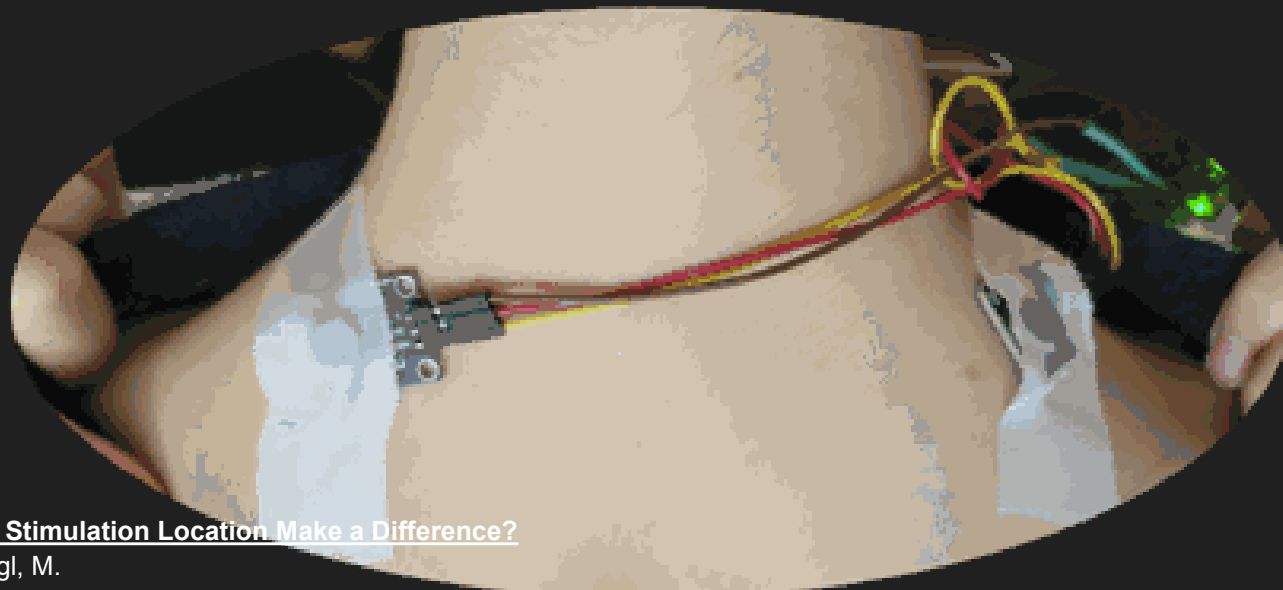
Experienced music players vs. novice

- *Experienced music players may perform worse than novice (non-significant),*
- Single-stimuli sensations are significantly more comfortable,
- No significant age relation,
- Stroking was rated as the most pleasant sensation,
- Subjective efficiency highest for tapping (>50%),
- Avg. Gwled game score almost the same for all 3 sensations.



What is Wakeful Rest?

- Wakeful Rest is an approach to bring you in a resting situation
- Using vibration motors (e.g. on the neck)
- Research questions: Does it work? How? Where?



Haptic Biofeedback for Wakeful Rest: Does Stimulation Location Make a Difference?

Lee, J.; Filpe, M.; Lepold, P.; Röddiger, T.; Beigl, M.

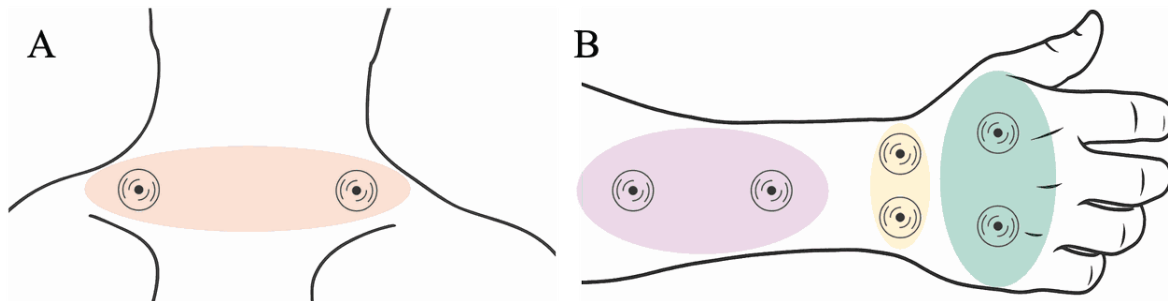
2025. Proceedings of the 2025 ACM International Symposium on Wearable Computers, 83–90, Association for Computing Machinery (ACM).

Body setup

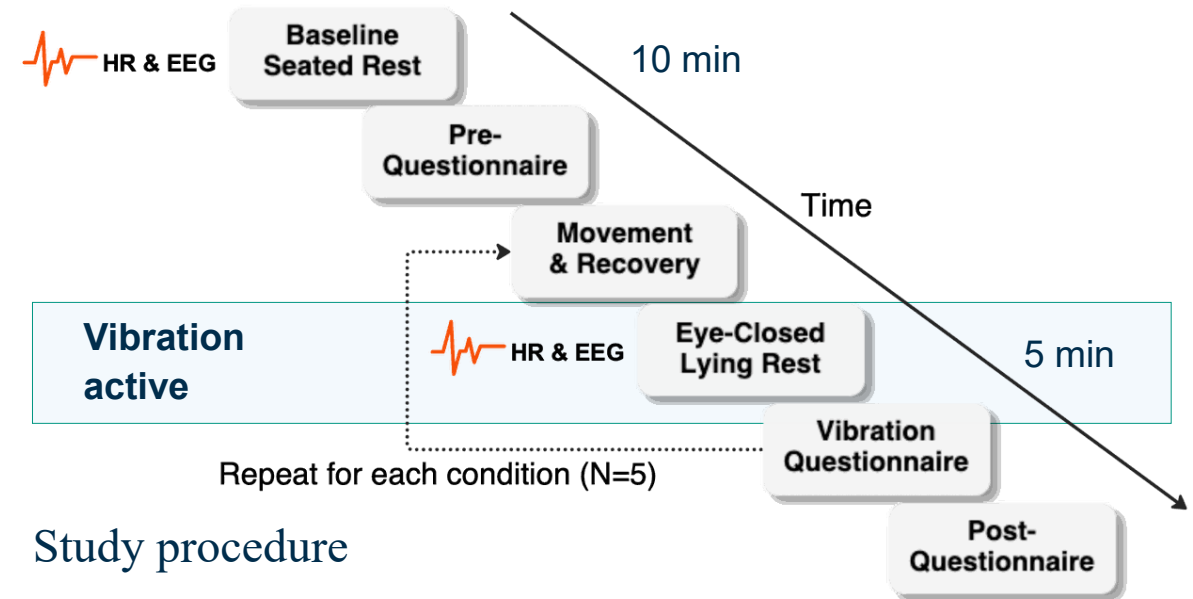
A major research question is on the body positions

Study: 20 participants (14 male, 6 female; $M_{age}=33.6$)

5 Conditions: 4 body positions + non-vibration



Locations of vibration on the body (A) Shoulder (B)
Right to Left: Forearm, Wrist, Hand



Measured Results

- There is significant difference in heart rate between baseline (no vibration) to wrist, hand, shoulder,
- Shoulder shows least variety among users,
- Due to the short time frame little absolute changes.
- Changes in alpha wave detection on the ear is too subtle to be significant

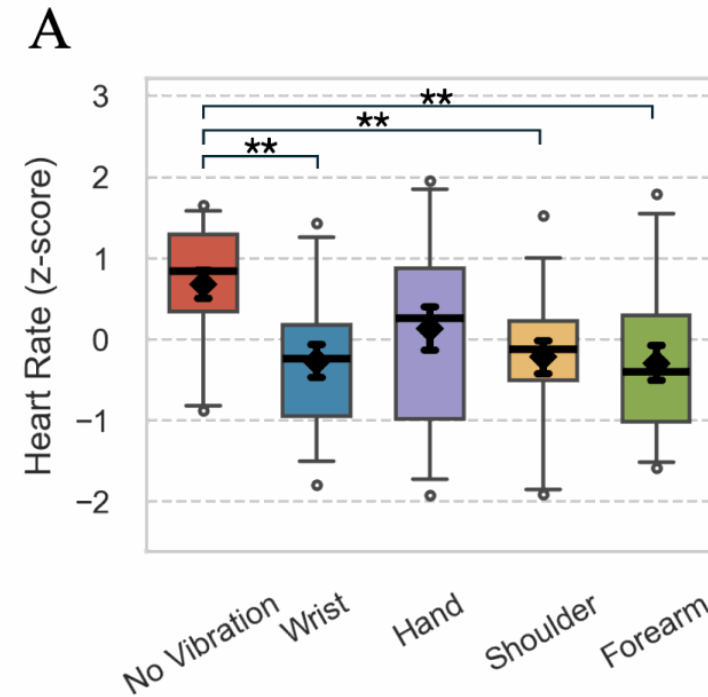


Fig A: Z-score normalized mean heart rate. Lower is more relaxed.

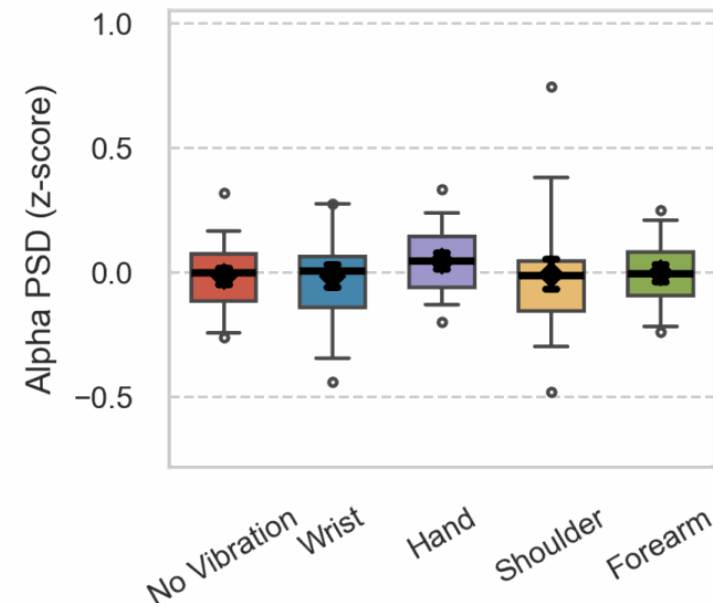


Fig B: Alpha PSD. Higher is more relaxed

Subjective Results

Subjective ratings show clear preferences for Shoulder and Forearm

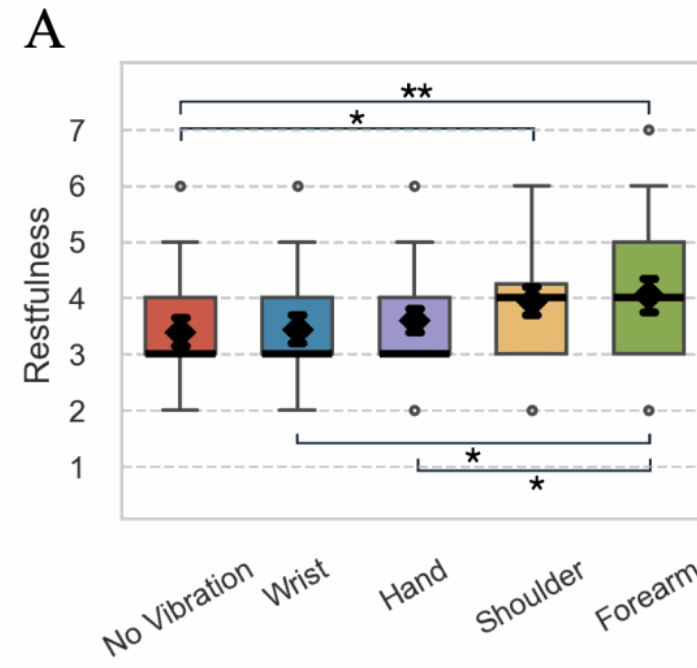


Fig A: Stanford Sleepiness Scale restful ratings

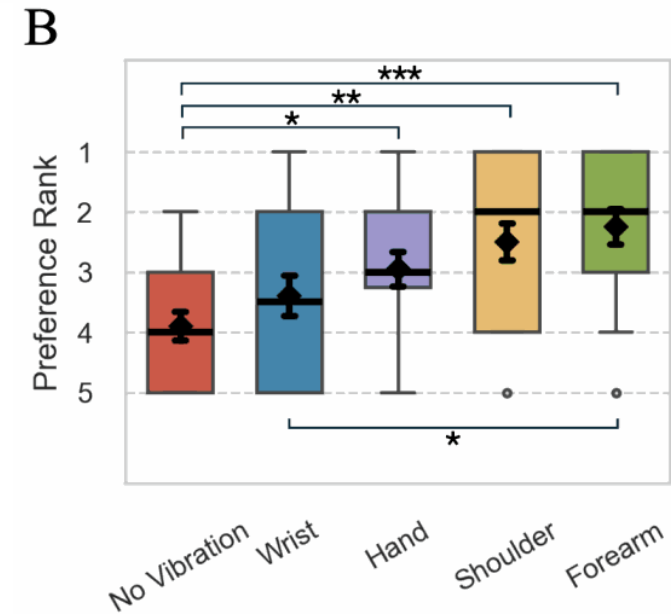


Fig B: Preference ranking (1=highest)